Kingdom of Saudi Arabia
Ministry of Higher Education
Imam Abdulrahman Bin Faisal University
College of Computer Sciences & Information Technology

جامعة الإمام عبدالرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

NEKHLAWI

نخلاوي

A project submitted
in partial fulfillment of the requirements for the degree of
*Bachelor of Science in {***Computer Information Sciences***}*

By

Zainab Hassan Al-Badi 2220002372
Hebah Ahmed Altheyab 2220001030
Doja Abdulmohsen Alnemer 2220002855
Zinab Mukhtar Alrashed 2220003194
Donia Abdullah Alfarhan 2220004173

Supervised by
Mrs. Albandary Mohammad Alamer

Committee Member Names
Dr. May Issa Aldossary
Mrs. Hina Gull

October 2025

DECLARATION

We hereby declare that this project report entitled Nekhlawi is a record of an original work done by us under the guidance of Supervisor Albandary Mohammad Alamer, Assistant Professor at College of CSIT, and that no part has been plagiarized without citations, which have been duly acknowledged. We also declare that it has not been previously or concurrently submitted for any other degree or award at any university or any other institutions. This project work is submitted in the partial fulfillment of the requirements for the degree of "Bachelor of Science in Computer Information Sciences" at Computer Information Sciences Department, College of Computer Sciences and Information Technology, Imam Abdulrahman Bin Faisal University.

Project Team:

Group Number: 6		
Team Member's Name	Signature	Date
Zainab Hassan Al-Badi		16/10/2025
Hebah Ahmed Altheyab		16/10/2025
Doja Abdulmohsen Alnemer		16/10/2025
Zinab Mukhtar Alrashed		16/10/2025
Donia Abdullah Alfarhan	Donia	16/10/2025

Project Supervisor:

Name	Signature	Date
Dr. Albandary Mohammad Alamer		16/10/2025

Table of Contents

LIST OF FIGURES 8

ACKNOWLEDGEMENT 9

ABSTRACT 10

CHAPTER 1: INTRODUCTION 11

1. Introduction 12

2. Problem Statement 12

3. Background and Review of Literature 14

3.1. Background 14

3.2. Methodological Approaches 14

3.2.1. Traditional and ML Methods 14

3.2.2. Deep and Hybrid Learning 14

3.2.2.1. Transfer Learning and Real-Time Systems 15

3.2.2.2. Application Domain: Palm and Fruit Studies 15

3.2.3. Palm-Specific Disease Detection 15

3.2.3.1. Date Fruit Grading / Maturity Estimation 16

3.3. Data Handling and Feature Engineering 17

3.3.1. Data Collection and Dataset Creation 17

3.3.2. Data Augmentation and Preprocessing 18

3.4. Model Evaluation and Comparative Analysis 19

3.5. Integration and Practical Applications 20

3.6. Research Gap and Recommendation 20

3.7. Literature Review Summary 23

4. Motivation 30

5. Justification 30

6. Aims & Objectives 31

7. Scope / Limitation of the Study 31

8. Significant of The Study 32

9. Professional and Ethical Implications 33

10. Project Organization 34

CHAPTER 2: SOFTWARE PROJECT MANAGEMENT PLAN (SPMP) 36

1. Overview 37

1.1. Project Summary 37

1.1.1. Purpose 37
1.1.2. Assumptions and Constraints 37
1.1.3. Project Deliverables 38
1.1.4. Master Schedule and Budget Summary 41
1.2. Evolution of the Plan 42
1.3. Document Structure 43
2. Definitions 44
3. Project Organization 45
3.1. External Interfaces 45
3.2. Internal Structure 46
3.2.1. The Project Manager 47
3.2.1.1. Scope of Responsibility 47
3.2.1.2. Internal Responsibilities 47
3.2.1.3. External Responsibilities 48
3.3. Project Roles and Responsibilities 48
4. Management Process 49
4.1. Start-up 50
4.1.1. Estimation 50
4.1.2. Staffing 50
4.1.3. Resource Acquisition 52
4.1.4. Staff Training 53
4.2. Work Planning 53
4.2.1. Work Activities 53
4.2.2. Schedule Allocation 54
4.2.3. Resource Allocation 57
4.2.4. Budget Allocation 58
4.3. Project Controls 58
4.3.1. Requirements Control 58
4.3.2. Schedule Control 59
4.3.2.1. Schedule Tracking 59
4.3.2.2. Schedule Performance Reports 59
4.3.2.3. Schedule Reviews 59
4.3.2.4. Progress Variance Monitoring 59
4.3.2.5. Progress Variance Resolution 59

4.3.2.6. Follow-up on Corrective Action 60
4.3.3. Budget Control 60
4.3.3.1. Cost Management 60
4.3.3.2. Methods to Ensure Cost Adherence 60
4.3.3.3. Cost Control 60
4.3.3.4. Contractor Cost Control 60
4.3.3.5. Cost Variance Measurement 60
4.3.3.6. Cost Variance Corrective Action 61
4.3.4. Quality Control 61
4.3.5. Project Reporting and Communication 61
4.3.5.1. Electronic Media 61
4.3.5.2. Meetings 61
4.3.5.3. Information Repository 61
4.3.5.4. Reviews 62
4.3.5.5. Status Reporting. 62
4.4. Risk Management 62
5. Technical Process 63
5.1. Process Model 63
5.2. Methods, Tools and Techniques 64
5.3. Project Infrastructure 64
5.4. Product Acceptance 65
6. Supporting Processes 66
6.1. Configuration Management 66
6.2. Independent Verification and Validation 66
6.3. Documentation 67
6.4. Quality Assurance 68
6.5. Reviews and Audits 68
6.6. Problem Resolution 68
6.7. Process Improvement 68
6.7.1. Systems/Software Process Improvement Lead 68
6.7.2. Systems Engineering Process Group 69
REFERENCES 70

LIST OF TABLES

Table 1: Research Gap and Recommendation 20
Table 2: Literature Review Summary 23
Table 3: Project Deliverables 38
Table 4: Table of Definitions 44
Table 5: Responsibility matrix 49
Table 6: Tools to be used and their cost 50
Table 7: Staff roles, skills and phases 51
Table 8: Resource Acquisition 52
Table 9: Resource Allocation 57
Table 10: Budget Allocation 58
Table 11: Project risk management plan 62

LIST OF FIGURES

Figure 1: Age of palm farmers in Al-Ahsa Oasis 13
Figure 2: Master Schedule 41
Figure 3: Project initial budget 42
Figure 4: External interface organizational chart 46
Figure 5: Organizational chart 47
Figure 6: Project Roles and Responsibilities 48
Figure 7: Work activities 53
Figure 8: Schedule Allocation 56
Figure 9: Waterfall Model Stages 64

ACKNOWLEDGEMENT

All praise and gratitude are due to Allah, first and last, for granting us the strength, knowledge, and perseverance, to Him belongs the grace and favor of every step we have taken and every idea we have achieved. We would like to express our sincere appreciation to Dr. Albandary Mohammad Alamer for her continuous guidance, valuable feedback, and patience throughout the project's development. She has been an outstanding mentor and a great source of encouragement throughout this journey. Our gratitude extends to Imam Abdulrahman Bin Faisal University for providing a stimulating academic environment and opening the doors of knowledge and experience, which have enabled us to reach this stage of achievement and success and serve our community. We are also thankful to our families and friends for their constant moral support, patience, and motivation during every stage of this journey.

Finally, we acknowledge the efforts of all individuals and organizations who contributed to this project, including agricultural experts and data providers who offered advice, shared knowledge, or provided assistance in improving the system's functionality and accuracy.

ABSTRACT

Date palms are considered an important agricultural resource in Saudi Arabia being a huge source for income and food security, however, it faces unnoticeable threats like how most of palm tree owners lack the knowledge to get proper care for their palms land diagnosing and curing diseases and pests, or even having general knowledge about the palm itself, making the only available solution is to get professional help which is mostly hard to get for small-scaled palm owners, in addition to the fact that most professionals are old people who their knowledge is at risk of vanishing in future years, so in this light this study proposes Nekhlawi app, which uses artificial intelligence, machine learning, and image processing and other technologies to offer the user multiple tools that helps with these issues, including Image scanning that functions as a diagnostic tool for pests and diseases and palms' general, a classification tool for date variants that links the users to a wiki that educates the user more about the palms and their dates. Moreover, it offers the ability to reserve an appointment with an expert to reach their assistance and professional services. The proposed solution will help enhance the health of trees, decrease the losses in agriculture, maintain cultural farming methods, and enable the non-expert users to have a reliable and readily accessible guidance. Through a blend of technology and local farming tradition, the application helps promote the sustainability of palm farming and the disconnection between the digital solutions and farming knowledge.

CHAPTER 1: INTRODUCTION

1. Introduction

According to UNESCO: "For centuries, many populations have been associated with the date palm tree, which has aided them in the construction of their civilizations in arid regions [1].

Representing prosperity, heritage, and sustainability, Date palms are vital agricultural and cultural resources in Saudi Arabia. They contribute significantly to food security, the industrial sector in production, trade, and the national economy [2].

However, many people, especially small-scale palm owners, such as households and community centers, often lack the required expertise and knowledge for proper tree care, pest management, and disease diagnosis. Where there's a gradual decline of traditional farming knowledge, usually held by older farmers, has made it difficult for new generations to access reliable guidance. This situation risks not only lower productivity and higher economic losses but also the erosion of cultural practices tied to palm cultivation [3].

To address this challenge, the project proposes a mobile application for palm care and disease, and pest identification. It utilizes computer vision, artificial intelligence and other technologies for palm disease and pest diagnosis, offers seasonal fertilizing and irrigation reminders, and includes a mini-wiki for knowledge acquisition. It also connects palm owners to agricultural experts through a consultation platform to offer safe advice while allowing experts to exchange their knowledge. By providing a useful, easy to access, and applicable solution, the project helps improve palm management and preservation of agricultural heritage for small-scale owners. In this report, problem background, project objectives, methodology, system design, implementation, and testing are outlined, followed by major findings, limitations, and recommendations for further work.

2. Problem Statement

Date palms are one of the most important crops in Saudi Arabia and the Gulf region, not only as a source of food and income, but also as a symbol of cultural identity and heritage. Evidence indicates that date palms have been present in the region since around 4000 BC. [4] Today, they play a pivotal role in national and regional economies through local consumption, exports, and tourism events such as date festivals. According to the National Center for Palms and Dates, Kingdom's date exports in 2024 amounted to more than 1.695 billion Saudi riyals [5].

Despite this importance, date palm owners—especially small farmers and home growers—face ongoing challenges in managing their trees effectively. Date palm cultivation is surrounded by a wide range of threats. Palm trees face major challenges, the most important of which are pests and diseases that significantly impact palm

production in the region, which, if left untreated, lead to reduced production, tree death, and economic losses equivalent to 28% of total production [6]. These threats include insect pests, fungal infections, and nutrient deficiencies. These problems are exacerbated by dy in the Al-Ahsa Oasis revealed that most farmers were over 40 years old, and their knowledge risks being lost in the absence of proper documentation or digital transfer mechanisms (see Figer 1) [7].

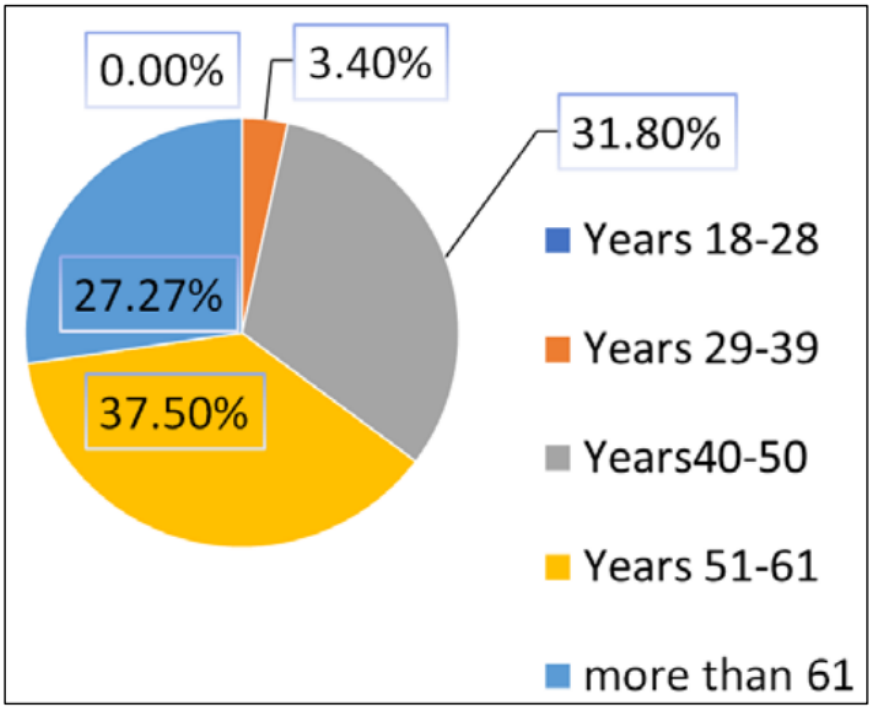

Figure 1: Age of palm farmers in Al-Ahsa Oasis

Current digital solutions in the Arab world remain limited and fragmented, primarily designed for large farms or government monitoring initiatives. Studies show that Arab agricultural applications are relatively few compared to their global counterparts, and that older or less educated farmers often struggle to adopt mobile technologies, facing barriers in language, ease of use, and digital literacy [8]. As a result, smallholders and inexperienced palm owners remain underserved and lack easy-to-use and accessible tools that meet their daily needs.

The absence of comprehensive digital platforms adapted to the Gulf countries widens this gap. Without interventions that combine modern diagnostic tools, practical guidance, and mechanisms for preserving specialized knowledge, the region risks losing agricultural competencies over time. This unresolved problem highlights the urgent need for solutions that link technological accessibility with agricultural sustainability.

3. Background and Review of Literature

3.1. Background

The Nekhlawi project will help alleviate a main issue that plagues the date palm industry in Saudi Arabia: an increasing generational knowledge loss that puts the economy and the culture at risk. The experience that goes into palm farming is mainly on the part of an aging generation, and the new and small-scale growers lack the expertise and equipment that will enable them to handle pests and diseases effectively, and this results in unnecessary agricultural losses. Nekhlawi application is one direct solution to this issue as a digital bridge is created, which democratizes expert knowledge. It has empowered growers to save their lives by using artificial intelligence to diagnose diseases and also give them a platform to consult experts. At the same time, it serves as an online library to store and share traditional cultivation techniques to protect this intangible cultural heritage. Consequently, the project is a critical intervention to enhance the economic sustainability of one of the main agricultural industries as well as to secure the survival of the date palm as a living cultural heritage.

3.2. Methodological Approaches

3.2.1. Traditional and ML Methods

The previous papers had used simple image processing (color thresholding, edge detection, K-means), and texture analysis (GLCM, LBP) to detect disease spots or rate date fruits. With the growth of digital data, handcrafted features were used in machine learning models, such as SVM, Random Forest, and KNN. The methods were easy and efficient in small-scale or binary detection but also needed manual extraction of features and were also prone to light and complicated data [9].

3.2.2. Deep and Hybrid Learning

Recent studies make use of deep CNNs (VGG, ResNet, DenseNet, Inception) that automatically extract hierarchical image features, and it is highly accurate in detecting multiple classes of diseases and in classifying fruit maturity. In other studies, networks are combined or hybrids of CNN + SVM are employed to perform ensemble prediction to enhance stability and generalization across difficult or unbalanced data [10] [11] [12] [13] [14].

3.2.2.1. Transfer Learning and Real-Time Systems

The use of transfer learning and pre-trained models (VGG19, Inception-ResNet, Xception) in the modern approaches helps minimize training time and data amounts. The lightweight CNNs such as DPXception allow real-time detection on mobile or drone systems, which are useful in farming precision and in practical field application [10].

3.2.2.2. Application Domain: Palm and Fruit Studies

Recent developments in deep learning and computer vision have significantly changed the field of date fruit grading and palm tree disease detection research by offering more scalable, effective, and objective evaluation techniques than manual methods. Modern algorithms now automatically identify disease symptoms and evaluate fruit maturity directly from image data, achieving high accuracy in real-world field scenarios, as opposed to depending only on expert visual inspection, which is laborious and subject to human error [10]. Interestingly, models like MobileNetV2 and DenseNet have outperformed previous machine learning techniques in the classification of important visual symptoms like brown spots and white scale. These models provide solutions that can be deployed on mobile devices for real-time field use [9]. Furthermore, technological integration extends beyond the visible spectrum; for example, thermal imaging and chlorophyll sensors allow for the early detection of infections and hidden pests before outward symptoms manifest [15] [16]. Large-scale grove monitoring is now possible thanks to remote sensing technologies, which enable targeted management and early intervention across extensive plantations [17]. Despite these significant advancements, the field continues to face significant methodological and practical obstacles that will be addressed in future work.

3.2.3. Palm-Specific Disease Detection

The automatic detection of diseases in palm trees has recently seen great strides made by the application of a combination of deep learning and machine learning techniques. One of the most remarkable networks is the Modified Residual Network (MResNet) which resulted in a significant validation accuracy of 100% on the strengthened datasets, especially when it came to spotting the brown spots and white scale issues [10]. Other diseases like bacterial leaf blight and leaf smut have also been detected by the MobileNet model, which had nearly 99.56% accuracy. This shows how powerful the deep convolutional architectures are when it comes to telling apart symptoms that could be similar [9]. Mobile and edge computing advancements made it possible for lightweight and powerful models to be used in the field. Among them are Effi-ShuffleNet and DPXception that are highly regarded for their computational efficiency and diagnostic accuracy [18]. In special cases like Sudden Decline Syndrome (SDS),

the hybrid algorithm FFMKO has united image enhancement with clustering techniques that result in a 94.2% accuracy and hence timely intervention [14]. Moreover, the machine learning frameworks employing support vector machines (SVM) for white scale disease stage classification have achieved an accuracy of 98.29%, thus facilitating the development of more specific treatment plans [19]. Most importantly, Research made it possible to detect fungal infections at an early stage by analyzing physiological indicators like chlorophyll content, while also improving significantly the detection of internal pests and regional stress with the help of thermal imaging and NDVI-based remote sensing, thus increasing accuracy and efficiency of management [20].

3.2.3.1. Date Fruit Grading / Maturity Estimation

The use of deep learning has played a significant role in the automation of post-harvest processes, especially in the grading and variety classification of date fruits. Deep learning has made a considerable leap in the quality of Convolutional Neural Networks, with the likes of Inception-ResNet-v2 and DenseNet121 with SE mechanisms already proving to be highly trustworthy with accuracies for quality classification exceeding 98% [21] [19]. For the task of variety identification, the YOLOv8 object detection model has surpassed all others by setting a new bar of 0.994 in mAP, thus demonstrating its suitability for large-scale sorting operations [22]. Maturity estimation, which is important for planning the harvesting process, is also conducted by deep models like ResNet that have achieved more than 99% accuracy in predicting the developmental stages [12]. The studies carried out on the VGG-16 architecture yielded very good results in this area as well, thus facilitating the robotic application process; for instance, the authors have reported up to 99.01% accuracy in type classification and 97.25% in maturity, respectively [18]. Furthermore, it is important to highlight that traditional image processing and machine learning techniques that emphasize measurable characteristics such as color, size, and shape still have an important role to play in conjunction with deep learning models; according to some reports, these traditional methods yield as high as 95% accuracy in practical grading [23]. Consequently, the combined models of logistic regression and artificial neural networks have been successful in identifying genetic varieties with an accuracy of 92.8% [24]. The availability of extensive datasets that are well-annotated, including images and videos recorded in natural orchard environments, has also been a major factor in research, and practical applications being slow to propagate. The resources are crucial for the training and validation of automated harvesting and grading systems, which is their major impact as well as the movement towards actual autonomous systems in precision agriculture [20].

Results:

The research deeds have been scrutinized in this paper point towards a remarkable and increasingly fast trend in the usage of advanced technology for both palm disease

detection and date fruit classification. Across this domain, various technologies ranging from conventional image processing to very complex deep learning systems and large-scale remote sensing are being applied successfully with consistently very high accuracy rates, often between 98% and 100% for specialized tasks. A crucial result of these studies is the non-invasive early detection that has been demonstrated as a feasibility which is not only the turning point from reactive treatment but also the integrated precision-based agricultural management that is made so much easier and cheaper. This evolution is heavily underpinned by the recent public datasets that have been created which represent the essential element of the model benchmarking and the future innovation lifts. Besides this, the practical applications have started to move forward as indicated by the in-field mobile diagnostic apps and foundational datasets created for robotic harvesting, which are very clearly and positively pointing towards the agricultural industry transformation. However, these significant achievements notwithstanding the major obstacle still lie in the need for the creation of systems that are not only highly accurate but also sufficiently robust, generalized, and cost-effective for reliable performance amidst the complexities and variations of real-world agriculture.

3.3. Data Handling and Feature Engineering

3.3.1. Data Collection and Dataset Creation

High-quality and diverse datasets are essential for building reliable models in palm disease and fruit taxonomy research. Researchers use multiple data collection methods and different tools to help them work efficiently. For example, in "Palm Leaf Health Management: A Hybrid Approach for Automated Disease Detection and Therapy Enhancement" researchers chose to capture high-resolution, multi-angle images of palm leaves using drones to document different disease stages and seasonal variations. This ensures that the datasets represent natural field variation rather than controlled laboratory conditions [13]. Similarly, other studies, such as "Date fruit detection dataset for automatic harvesting" used drone photography to collect red, green, and blue color images and videos, captured under varying lighting conditions between 6 a.m. and 6 p.m., reflecting realistic challenges, such as obstruction and size variation [20]. On the other hand, collecting data from the ground remains equally important. Another team, "Dataset of infected date palm leaves for palm tree disease detection and classification" collected live images from ten farms in Medina, Saudi Arabia, using smartphones and DSLR cameras to capture real disease symptoms on leaves affected by fungal, physiological, and pest-induced disorders [25]. Furthermore, the project "IHDS: Intelligent Harvesting Decision System for Date Fruit Based on Maturity Stage Using Deep Learning and Computer Vision," collected field data using DSLR cameras in palm groves under varying natural lighting conditions [11]. The images were manually classified according to seven maturity stages for deep learning training. Similarly,

"DPXception: A Lightweight CNN for Image-Based Date Palm Species Classification" did [15].

Collectively, these efforts demonstrate the effectiveness of deep learning in agricultural image analysis. Diverse, multi-source datasets whether captured by drones, handheld cameras, or natural-environment imaging for species classification form the experimental foundation for developing high-performance, field-adapted deep learning models in smart agriculture. Regardless of the chosen data collection method, ensuring data quality and diversity remains essential for achieving reliable, generalizable, and scalable models that can support precision-farming applications.

3.3.2. Data Augmentation and Preprocessing

Many researchers resort to data augmentation and preprocessing in order to make the data more accurate, consistent, and understandable for the model, as these processes clean and prepare the data before feeding it into the model. In addition, they increase the size of the dataset by creating modified copies without the need to collect new data. In the study "Palm Tree Disease Detection and Classification Using Residual Network and Transfer Learning of Inception ResNet," the authors applied comprehensive preprocessing that included image normalization, resizing, and extensive data augmentation specifically flipping, rotation, and brightness adjustment—to enhance the variability of the dataset. This step increased the diversity of visual inputs and enabled the modified ResNet model to reach 100% accuracy after augmentation, proving the direct impact of such preprocessing on model performance [10]. Similarly, "A Machine Learning-Based Framework for Stage-Wise Classification of Date Palm White Scale Disease" performed contrast enhancement and filtering prior to feature extraction, allowing the model to detect subtle texture differences between infection stages while reducing image noise and lighting bias [19].

Furthermore, the project "Plant Leaf Disease Detection Using Computer Vision and Machine Learning Algorithms" implemented a classical multi-step preprocessing pipeline combining histogram equalization for contrast correction, K-means clustering for lesion segmentation, and Principal Component Analysis (PCA) for dimensionality reduction [25]. This hybrid strategy reduced computational load while preserving the most distinctive visual features, directly enhancing the classification accuracy of CNN and SVM models. Beyond conventional transformations, the study "Advances and Challenges in Computer Vision for Image-Based Plant Disease Detection" discussed the growing trend of using deep learning-based preprocessing, where convolutional filters automatically learn to suppress background noise and normalize illumination during training. The same study emphasized the potential to mitigate data imbalance through modern augmentation strategies such as color jittering, zooming, rotation, and noise injection, which simulate real-world variability and prevent overfitting [27].

Overall, preprocessing and augmentation techniques demonstrate that effective data preparation is a fundamental prerequisite for developing high-performance deep learning systems in palm disease detection and agricultural image analysis.

3.4. Model Evaluation and Comparative Analysis

In order to integrate artificial intelligence (AI) in agriculture, models must be evaluated using various metrics such as accuracy, F1-score, recall, precision, and confusion matrix. However, these models can be used to detect diseases or classify fruits.
Two types of learning methodologies were evaluated and used to classify White spot disease:

1. Classical machine learning: using support vector machine (SVM), k-nearest neighbors (KNN)
2. Ensemble learning: using random forest (RF) and light gradient boosting machine (LightGBM)

The study states that SVM outperformed other models by an accuracy rate of 98.29% [19]. While another study showed that it achieved 92.8% only classifying the Red Palm Weevil pest [15]. Another well-known model is Convolutional Neural Network (CNN), which was used to detect differences between Leaf Spots and Blight Spots diseases with an accuracy ratio of success rate equal to 97.9% [15]. CNN different architectures (Inception-v3, Inception-ResNet-v2, VGG19) are also used to categories date fruit into accepted and rejected samples to create a dataset for Medjool, Saeidi, and El-Wadi date fruit, among these different architectures, Inception-ResNet-v2 showed the best performance with an accuracy of 98.99% which increases the ability of utilizing it to develop a suitable computer vision system [21]. However, combining both classifiers SVM and CNN along with RGB image processing, segmentation and feature extraction can score accuracy rates of 93%, 94.5% and 96.3 % for classifying the high class, classifying middle class, and classifying low class respectively which can be furthermore increased up to 100% if used tools to detect date's skin are different [22]. When comparing ResNet (Residual Network), MresNet (Modified Residual Network), and IncResNet (Inception-Residual Network) to detect healthy palms or those infected with white scale or brown spots palms, the study shows that MresNet outperformed other models with accuracy rate of 99.62% [10]. Models can be further improved and utilized as study shows that DPXception model, which is driven from using only the first 100 layers of Xception, making it more efficient and lightweight, secondly, the model applies augmentation and normalization to improve accuracy and robustness of it along with other utilizations, the DPXception model outperformed seven well known models in term of F1-score, accuracy, and inference time [18]. In conclusion, although the different models showed a good performance in terms of other majors, there are still some challenges that have not been solved yet, such as the need for a huge amount of

data, the high cost for data labelling and obtaining data from hard-to-reach places remaining hard to solve [28].

3.5. Integration and Practical Applications

Agriculture is an important traditional field and integrating it with modern technologies such as deep learning and computer vision will help move it forward through precision farming and sustainability. Moreover, it helps manage palm health throughout disease detection and enhancing therapy using image processing and neural networks [13]. To help overcome human errors and limitations, automated gardening systems are used for earlier classification through sensor-based techniques and statistical algorithms, raising accuracy and enabling robotic harvesting by deep learning [22]. ResNet and VGG-16 are embedded in harvesting robots to perform tasks such as locating fruit and monitoring their quality. To do further estimation, drones are used to provide the system with pictures along with multiple cameras set up [12]. Combining CNN and SVM in an intelligent hybrid detection system can help identify palm issues such as Leaf and Blight spots, making it possible to provide treatment to the palm faster [15]. The evolution of smart agriculture has led to the development of an intelligent harvesting decision system (IHDS) that assesses the maturity level of date fruit using computer vision and deep learning, making smart harvesting decisions which will improve the quality of the yield [11]. Studies also showed that integrating the internet of things (IoT) along with deep learning algorithms will improve quality and reduce cost and labor, supporting sustainability in agriculture by optimizing usage of [26]. Finally, although integrating new technologies can improve agriculture, the deployment cost is high, and the models are trained on limited data, so they are not generalized. There is also a lack of explanation of how the model makes certain decisions, but smart agriculture is continuously improving.

3.6. Research Gap and Recommendation

Table 1: Research Gap and Recommendation

Category	Gap	Recommendation
Methodological Approaches	Many models focus on binary classification rather than multi-disease or multi-maturity stage classification.	Employ advanced sampling (SMOTE, data augmentation) and cost-sensitive learning methods and scale classification tasks to multi-class and multi-label scenarios.

Category	Gap	Recommendation
	Most of the experiments are based in laboratories under controlled lighting, and few of them prove the models on real farm pictures taken with drones or mobile devices.	Apply real-world experiments and try to include noise in training datasets to imitate real-world noise
	Most studies offer limited insight into how the model differentiates between similar symptoms.	Use explainable AI frameworks and statistics to assist farmers and agronomists in understanding the diagnosis.
	Most research focuses on detection accuracy without connecting the output to something actionable like treatment recommendations.	Use the output to make something clear, meaningful and usable for the end user.
	CNNs and their variants require high computational power, limiting the ability use it on mobile devices.	Integrate image detection with agronomic knowledge bases to automatically recommend treatments and can also develop interfaces where farmers will upload images and receive immediate feedback.
	Current models handle images at once instead of monitoring disease progress or maturation of fruits.	Use spatio-temporal deep networks (e.g., CNN-LSTM) for tracking disease evolution.
Application Domain: Palm and Fruit Studies	Few public image collections depicting different palm diseases of good quality are available.	Similar to the recent general disease dataset effort [25], create and share large, annotated datasets covering more diseases, palm types, and regions.
	The models that perform excellently in labs usually get unsuccessful results in actual farms with changeable lighting and backgrounds.	Through training on more varied, field data, real-world condition tolerant models will be developed.
	Highly sophisticated devices such as thermal cameras are not only expensive but also difficult to use for most farmers.	Shift attention to the development of inexpensive and easy-to-use tools like smartphone apps to make the technology usable in the field and accessible to farmers [15].

Category	Gap	Recommendation
	Most models are too specialized, and they can only detect one disease at a time.	Develop systems that are more comprehensive, by merging different data types (for instance, visual, thermal, and satellite) to identify multiple issues simultaneously.
	Lack of datasets emphasizing defective or rejected dates.	Quality control models stand to benefit a lot from new datasets comprising a vast and properly labeled assortment of typical fruit defects.
	The date industry still depends on the slow, inconsistent manual sorting process to a large extent.	Encourage the adoption of affordable and user-friendly automated systems that are able to demonstrate clear financial benefits over manual labor.
	The absence of real-world orchard data (images and videos) has been the major constraint for the development of harvesting robots.	Build massive public datasets comprising images and videos of dates in natural environments for the purpose of training and evaluating robotic systems [20].
	Natural variations among dates (type, ripeness, size) make it hard for just one model to classify them all accurately.	Apply cutting-edge models such as object detectors (like YOLOv8), which are purposely made to manage the most minute visual differences.
Model Evaluation and Comparative Analysis	Not Enough Good Data: There are not enough excellent, real-world images depicting date palm diseases for the proper training of AI models. The process of gathering this data is both costly and laborious.	Create a Public Photo Library: This step could be a great way to put an end to the endless and tedious arguments over the rightness of the AI models involved as all parties will be able to figure out who is the best one by simply observing the results of bearing the same images through AI.
	Models Don't Always Work in the Real World: An AI model trained in one area may not give good results in another area because of differences in lighting, weather, or palm varieties. The issue of reliability across regions is therefore brought up	Develop Smarter, Lighter AI: This should be a priority since the AI models will be simpler, will consume less power, and hence, can be used on devices of lower capacity like those of a smartphone.

Category	Gap	Recommendation
	The prices of drones, high-end cameras, and powerful computers are such that only a few farmers can afford and use these advanced AI systems.	Focus on Complete, Affordable Systems: AI technology would be made virtually non-restrictive only if researchers provided the total package of the technology, that is, all the way from the application of low-cost drones or phone cameras for capturing images to presenting a simple app that performs diagnosis. This way the technology will not only be ready for the scholars' laboratories but also for the daily farmers in the field.
	The majority of AI models are not capable of giving reasons for the diagnosis they rendered (for instance, which point on a leaf signaled the presence of a disease). Consequently, farmers find it hard to put their trust in the technology.	Build Trust with Explainable AI (XAI): The future of AI would be one that can justify its actions and give reasons for its choices. The same way, for example, the app could spot a leaf portion showing disease and highlight it for the farmer to see what the app sees.

3.7. Literature Review Summary

Table 2: Literature Review Summary

No.	Title	Authors	Year	Aim	Methodology	Results
1	Palm Leaf Health Management: A Hybrid Approach for Automated Disease Detection and Therapy Enhancement	S. M. Nuruzzaman Nobel, Md. Asif Imran, Nahida Zaman Bina, Md. Mohsin Kabir, Mejdl Safran, Sultan Alfarhood, and M. F. Mridha	2024	Develop a hybrid deep-learning model for precise and scalable detection of palm leaf diseases.	Combined ECA-Net, ResNet50, and DenseNet201 with transfer learning; dataset captured via drones and augmented with rotation, color adjustment, and noise injection.	> 99 % training accuracy and > 98 % validation accuracy [13].
2	External Features-Based Approach to Date Grading and Analysis with Image Processing	Shabana Habib, Ishrat Khan, Suliman Aladhadh, Muhammad Islam and Sheroz Khan	2022	Automate grading of date fruits based on visual appearance to replace manual inspection.	Applied RGB image processing, segmentation, and feature extraction (color, texture, shape, size) with ML classifiers (SVM, neural networks).	accuracy of 93% was achieved for classifying the high class, 94.5% for classifying the middle class, and 96.3% for classifying the low class [23].

No.	Title	Authors	Year	Aim	Methodology	Results
3	Deep Learning and Computer Vision for Estimating Date Fruits Type, Maturity Level, and Weight	Mohammed Faisal, Fahad Albogamy, Hebah Elgibreen, Mohammed Algabri and Fattoh Abdu Alqershi	2020	Build a unified vision system to estimate date type, maturity, and weight.	CNNs (VGG-16, AlexNet, ResNet) for maturity/type classification; regression CNNs for weight estimation; data collected in orchards under varied light.	DTES achieved maximum performance of 99.175% accuracy, 99.225% F1 score, 99.8% average precision, and 99.05% average recall, DMES achieved a performance of 99.058% accuracy, 99.34% F1 score, 99.64% average precision, and 99.08% average recall, and DWES were 84.27% SVM-linear [12].
4	Date Fruit Detection and Classification Based on Its Variety Using Deep Learning Technology	Ahad Almutairi, Jawza Alharbi, Shouq Alharbi, Haifa F. Alhasson, Shuaa S. Alharbi and Shabana Habib	2024	Identify different date varieties automatically using modern deep learning models.	YOLOv5, YOLOv7 and YOLOv8 detectors and CNNs on 1,750 images across 8 – 9 Saudi varieties; included data augmentation and transfer learning.	YOLOv8 achieves a mean Recall of 0.99 % and Precision of 0.991% with an Intersection over Union (IoU) colorred in the range of [0-1], a mean Average Precision of 0.994%, a mean Average Precision of [0.5:0.95] and a mean Average Precision of [0.5:0.85] [22].
5	Palm tree disease detection and classification using residual network and transfer learning of inception ResNet	Mostafa Ahmed and Ali Ahmed	2023	Evaluate and compare deep residual models for palm disease detection.	Tested ResNet, Modified ResNet (MResNet), and Inception-ResNet on 2,631 labeled palm leaf images with and without augmentation.	Accuracy of 99.62 % (original) and 100 % (augmented) [10].
6	A Machine Learning Based Framework for a Stage-Wise Classification of Date Palm White Scale Disease	Abdelaaziz Hessane, Ahmed El Youssefi, Yousef Farhaoui, Badraddine Aghoutane, and Fatima Amounas	2023	Classify white-scale disease in palm leaves across different infection stages.	Extracted 80 GLCM texture + 9 HSV color features; compared SVM, KNN, RF, LightGBM classifiers.	SVM (GLCM + HSV) achieved 98.29% accuracy [19].
7	Plant leaf disease detection using computer vision and machine learning algorithms	Sunil S. Harakannanavar, Jayashri M. Rudagi b, Veena I Puranikmathb, Ayesha Siddiquaa and R Pramodhini	2022	Create a hybrid model for tomato leaf disease detection.	Enhanced and segmented images (HE + K-Means + Contour tracing); extracted DWT, PCA, GLCM features; classified with SVM, KNN, CNN.	CNN with hybrid features achieved accuracy of > 95% [29].

No.	Title	Authors	Year	Aim	Methodology	Results
8	Classification of Palm Trees Diseases using Convolutional Neural Network	Marwan Abu-zanona1, Said Elaiwat2, Shayma'a Younis3, Nisreen Innab4, M. M. Kamruzzaman	2022	Design a lightweight CNN for multi-disease palm classification.	4-layer CNN and benchmarked against MobileNet and VGG-16 using the DP-Disease dataset (4 diseases + healthy).	Achieved accuracy of 99.10% in CNN, 99.35% in VGG and 99.56% in MobileNet [9].
9	An Intelligent Approach for Detecting Palm Trees Diseases using Image Processing and Machine Learning	Hazem Alaa, Khaled Waleed, Moataz Samir, Mohamed Tarek, Hager Sobeah and Mustafa Abdul Salam	2020	Combine visual and thermal imaging for comprehensive palm disease detection.	CNN (VGG-16) on RGB leaf images for external diseases; SVM (GLCM on thermal images) for Red Palm Weevil.	Achieved accuracy of 97.9% in CNN and 92.8% in SVM [15].
10	A new hybrid algorithm for intelligent detection of sudden decline syndrome of date palm disease	Aurangzeb Magsi, JavedAhmed Mahar, Abdullah Maitlo, MuneerAhmad, MirzaAbdur Razzaq, MohammadArif Sobhan Bhuiyan and Teh JiaYew	2023	Detect SDS in date palms using a computationally efficient hybrid model.	Fuzzy Fast Multi-Otsu K-Means (FFMKO) combining fuzzy filtering, Otsu thresholding, and K-Means segmentation; LBP + RGB features extracted.	Achieved accuracy of 94.17% [14].
11	Palm Tree Diseases Detection Using Deep Learning: A Short Review	Rechachi Bouthaina, Rezeg Khaled, Kazar Okba and Merizig Abdelhak	2024	Review deep-learning techniques applied to palm tree disease identification.	Compared CNN architectures (VGG, ResNet, DenseNet, MobileNet) and hybrid DL–ML models; discussed datasets and IoT links.	Achieved accuracy of 92%–100% across studies [26].
12	Advances and Challenges in Computer Vision for Image-Based Plant Disease Detection: A Comprehensive Survey of Machine and Deep Learning Approaches	Syed Asif Ahmad Qadri, Nen-Fu Huang, Taiba Majid Wani and Showkat Ahmad Bhat	2024	Survey computer-vision pipelines and benchmark ML vs DL models in agriculture.	Reviewed workflows (data capture, segmentation, feature extraction, classification); analyzed EfficientNetV2, VGG, RF, SVM performance.	CNNs achieved > 99% [27].

No.	Title	Authors	Year	Aim	Methodology	Results
13	IHDS: Intelligent Harvesting Decision System for Date Fruit Based on Maturity Stage Using Deep Learning and Computer Vision	Mohammed Faisal, Mansour Alsulaiman, Mohammed Arafah and Mohamed Amine Mekhtiche	2020	Automate maturity-stage detection to guide harvesting decisions.	VGG-19, Inception-v3, NASNet	Achieved accuracy of 99.7% using VGG-19 [11].
14	A Deep Learning Approach to Classification of Diseases in Date Palm Leaves	Sameera V Mohd Sagheer, Orwel P V, P M Ameer, Amal BaQais and Shaeen Kalathil	2025	Develop an automated deep-learning system to classify multiple date-palm leaf diseases with high precision.	Used pretrained CNN architectures (InceptionV3, DenseNet, MobileNet) with transfer learning.	DenseNet and MobileNet, achieved accuracies of > 95%. MobileNetV2 achieving accuracy of 96.99% and macro-average F1-score of 0.97 [30].
15	DPXception: a lightweight CNN for image-based date palm species classi cation	Mejdl Safran, Waleed Alrajhi and Sultan Alfarhood	2024	Introduce a new dataset and a new model for image-based date palm species classification	Data augmentation techniques, Developed a lightweight CNN architecture called DPXception (Date Palm Xception).	Achieved accuracy of 92.9%, F1-score of 93% and inference time of 0.0513 seconds [18].
16	Deep Learning for Image-Based Plant Growth Monitoring: A Review	Kin Sam Yen	2022	Review modern DL applications for monitoring plant growth via images.	Surveyed post-2017 studies using CNNs, YOLO, Mask R-CNN, and regression models; compared tasks of classification, detection, and temporal growth estimation.	Showed DL models' superiority in automatic feature extraction but noted challenges data-hungriness, annotation cost, and difficulty capturing spatial-temporal features [28].
17	Individual Palm Tree Detection Using Deep Learning on RGB Imagery to Support Tree Inventory	María Culman, Stephanie Delalieux and Kristof Van Tricht	2020	Create the first large cale spatial inventory of Phoenix dactylifera and Phoenix canariensis palms in Alicante, Spain, using deep learning and high resolution aerial imagery.	CNN-based detection model (RetinaNet).	Achieved a mean average precision (mAP) of 0.861 [31].

No.	Title	Authors	Year	Aim	Methodology	Results
18	Plant growth information measurement based on object detection and image fusion using a smart farm robot	Songhee Cho, Taehyeong Kim, Dae-Hyun Jung, Soo Hyun Park, Yunseong Na, Yong Seok Ihn and KangGeon Kim a	2023	Develop a smart farm robot that can measure plant growth information which to overcome challenges in traditional manual measures especially where plants look similar	RGB-D depth camera (Intel RealSense L515), depth-based pixel counting to measure Stem diameter, YOLO to identify flowers and branch points, LiDAR to measure crop height	The RGB-D dataset improved mean average precision mAP to 92.3% for Tomatoes, 89.6% for Cucumbers. This represents a 3.8% improvement for tomatoes and 0.8% improvement for cucumbers over RGB-only detection [32].
19	Smart farming technologies and changes to farm work: New insights into on-farm experiences	Ruth Nettle and Julie Ingram	2025	Examine how Ai driven Smart Farming Technologies (SFT) are changing the nature of farm work in the UK and Australia.	Drones and NDVI imaging, (SFTs) such as robotics, data systems, sensors, and automation, 17 farmers (10 from the UK and 7 from Australia), (30 participants including advisers and industry reps) was used to validate findings.	SFTs reduced physical work, increased data handling, higher digital skills demand, mixed impacts on well-being, and autonomy [33].
20	Sustainable deep vision systems to estimate the quality of date fruits through attention-enhanced deep learning systems	Esraa Hassan, Sarah Abu Ghazalah, Nora El-Rashidy, Tarek Abd El-Hafeez and Mahmoud Y. Shams	2025	Create robust attention-enhanced deep models for accurate date fruit quality classification under real-farm conditions.	Integrating a DenseNet121 model pre-trained on ImageNet with a Squeeze-and-Excitation (SE).	DenseNet121+SE achieved accuracy of 98.25%, 98.02% precision, 97.02% recall and a 97.49% F1-score. YOLOv8n achieved accuracy of 96.04%, 99.76% precision, 99.7% recall and a 99.73% F1- score [24].

No.	Title	Authors	Year	Aim	Methodology	Results
21	Advancements in Detecting Spot Fungal Diseases on date palm leaves: Development and validation of chlorophyll-based detection models	Khaled Hussein Arafat	2024	Create and confirm new models based on chlorophyll that would enable the fast, reliable, and non-invasive detection of fungal spot diseases on date palm leaves, thus, essentially changing the way diseases are managed.	The seedlings of date palms were subjected to pathogenic fungi, and researchers were assessing the disease severity (DS %) and the related leaf chlorophyll content over time.	The researchers discovered that there was a strong negative correlation between chlorophyll and disease severity. A statistical model that relied on chlorophyll content was able to effectively predict the level of severity of the disease, accounting for up to 75.84% of its variation (R^2=75.84%$) [16]
22	Dataset of infected date palm leaves for palm tree disease detection and classification	Abdallah Namoun, Ahmad B. Alkhodre, Adnan Ahmad Abi Sen, Yazed Alsaawy and Hani Almoamari	2024	Provide a large, public dataset to support DL-based disease detection.	Collected 608 raw images and processed them to 3,089 RGB images (300 × 300 px) covering 8 leaf disease classes and healthy samples from 10 Saudi farms.	Released the first comprehensive multi-class palm-leaf dataset, enabling researchers to benchmark and train DL models for real-world diagnosis [25].
23	Classification of dates quality with the help of deep learning technology using captured images	Waleed Kamel ElHelew, Dalia Abo Bakr, Sahar Zayan and Muhammad Ahmad Mayhoub	2024	Classify date fruit quality (accepted or rejected) with machine learning technology to reduce cost, time, and improve the quality of final product.	Used transfer learning with Inception-v3, Inception-ResNet-v2, and VGG19.	Inception-ResNet-v2 achieved accuracy of 98.99% [21].
24	Early Detection of Date palm Diseases in the New Valley Governorate of NDVI-based Digital Cartographic Modeling	Khaled Hussein Arafat, Mohammed HA Hassan, Wael M. Al-Metwaly and Wesam H. Mohammed	2025	Detect palm disease hotspots early.	Integrated multi-temporal Landsat NDVI data (2005–2024) with GIS and field validation to map stress zones in Egypt's New Valley Governorate.	achieved accuracy of 87.3% [17].

No.	Title	Authors	Year	Aim	Methodology	Results
25	Date fruit detection dataset for automatic harvesting	Yousra Zarouit, Hassan Zekkouri, Mohamed Ouhda b and Brahim Aksasse	2023	Build a realistic dataset to train object-detection models for robotic date harvesting.	Collected 9,092 RGB images/videos of 4 Moroccan varieties with diverse lighting/occlusions; annotated in YOLO format for detection, variety, and maturity.	Produced the largest open dataset for date harvesting covering detection + classification + maturity supporting robust computer-vision model development [20].
26	An Economic Study: Date Palm Farmers in Al Ahsa Oasis-Kingdom of Saudi Arabia	Ghada Alkhaldi, Abda Emam and Ezzeddine Mosbah	2022	Discover the factors impacting the production and profits of date palmfarmers in Al Ahsa Oasis in K ingdom of Saudi Arabia for seasons 2020/2021.	Data analysis was conducted through descriptive statistics for economic and social variables, regression analysis, and Likert scale.	Date palm farmers are aged 51–61 (37%), married (95.4%), and have over 10 years of farming experience (81.82%). 48.86% of date palm farmers earn about 10,000 SAR [34].
27	Automated Feature-Specific Tree Species Identification from Natural Images Using Deep Semi-Supervised Learning	Dewald Homana and Johan A. du Preez	2021	Recognize tree species from natural images using minimal labeled data.	Combined EfficientNet-B0 with FixMatch semi-supervised.	Achieved accuracy of 94.04% top 5 for leaves and 83.04% top 5 accuracy for bark [35].
28	Medicinal Plant Recognition Based on the Seedling Image and Deep Learning	Khalid M. O. Nahar, Nesrine Atitallah, Mohammed M. Abu Shquier, Dalal Z. Zreiqat and Khaled M. Alhawity4	2024	Identify medicinal plants at seedling stage for conservation and agriculture.	Inception 3, a deep convolutional neural network (CNN) architecture	Achieved accuracy of > 96%–98% across classes [36].
29	Deep-Learning-Based Automated Palm Tree Counting and Geolocation in Large Farms from Aerial Geotagged Images	Adel Ammar and Anis Koubaa	2021	Automate counting & mapping of palms using UAV images.	Trained Faster R-CNN on 217 drone images (9,873 instances); added custom geolocation algorithm converting detections to GPS coords.	Achieved a precision of 94%, recall of 84% and an average precision (AP) of 0.5 attains 83% [37].

No.	Title	Authors	Year	Aim	Methodology	Results
30	A Review of Plant Phenotypic Image Recognition Technology Based on Deep Learning	Jianbin Xiong, Dezheng Yu, Shuangyin Liu, Lei Shu, Xiaochan Wang and Zhaoke Liu	2021	Review deep-learning applications for plant-trait recognition (color, texture, shape, disease).	Compared CNN, DBN, RNN, and SAE models across classification, segmentation, and temporal analyses.	CNNs achieved < 99% accuracy and dominated PPIR tasks; DBNs and RNNs handled complex or time-series data [38].
31	A Study on the Economic Feasibility of Date Palm Cultivation in the Al-Hassa Oasis of Saudi Arabia	A. Al-Abbad, M. Al-Jamal, Z. Al-Elaiw, F. Al-Shreed and H. Belaifa	2018	Evaluate profitability and sustainability of Al-Hassa date-palm farming.	Cost, income, and SWOT analysis using NDPRC field data.	The study showed that palm cultivation in the Al-Ahsa Oasis is economically viable, with an average net income of 5,800 riyals per hectare, with a production of 48 kg of dates per palm tree and a price of 4 riyals per kilogram [39].

4. Motivation

Palm trees hold deep value in terms of culture, economy, and agricultural importance, particularly in the Gulf region, where date farming has been our foundation and our roots since the time of our ancestors. However, the person who is interested in planting and gardening palm trees often struggles to maintain tree health due to limited expert knowledge. Much of this knowledge is held by ageing farmers without proper technology to transfer this knowledge, and there is a risk of losing it. Moreover, palm diseases can lead to considerable economic losses for farmers and communities. We are motivated to develop a mobile application that uses AI and modern technologies to make palm and date care easy, reliable and sustainable. This idea combines our passion for technology and keeps agricultural tradition, supports local farmers and secures the future of palm farming.

5. Justification

Several factors are illustrated below to offer justification for the main reasons behind this project:

1. Knowledge safekeeping: Helps to save the expertise of aging farmers and transfer their knowledge into modern, accessible platform.
2. Accessibility: Provide palm owners with easy and reliable diagnosis and expert consultation.
3. Economic value: Reduces losses caused by palm diseases through timely detection and treatment guidance.

4. Cultural importance: Protect and support the rich tradition of palm and date farming to ensure sustainability for the next generations.
5. Tangible benefit: Offers recommendations, reminders and other services to improve palm management.

6. Aims & Objectives

The aim of this project is to utilize modern technologies to develop an application that scans and diagnoses palms and dates, providing customers with accurate information about diseases or nutrition, and helping maintain palm health via consultations and the mini wiki. Moreover, it supports our rich farming traditions through digital transformation in the agriculture field.

To achieve this aim, the project seeks to:

- Develop an AI-powered palm tree care tool that is expected to detect at least 70% of common palms health issues.
- Provides medicine recommendations that aim to reduce farmers' diagnosis time by at least 30%.
- Within 3 months, establish a support system that connects clients with palm farmers experts through chat, video calls and on-site visits, targeting an average response time under 24 hours for chat and video requests.
- Publish at least 25 educational resources (articles, infographics, and tutorials) on the mini-wiki platform.
- Reduce palm health losses by an expected margin of 30% or more.
- Protect and promote the rich tradition of palm farming by providing easy access to information by the end of 2026.

7. Scope / Limitation of the Study

The project is designed to support small-scale palm owners in Saudi Arabia, specifically individuals and institutions who have limited experience in palm cultivation, such as private households, residential compounds, and community centers.

The mobile application will provide the following functionalities:

- AI-based disease and pest diagnosis through image recognition.
- Identification of Date Types and their key characteristics.
- Consultation from experts via online or on-site paid services.
- Reminders and notifications for suggested fertilization, irrigation, and pruning schedules.
- Awareness and education content through a mini-wiki about palm trees, their types, and proper care practices.

Excluded from the project scope are broader agricultural domains, including large-scale farm management, ERP-style agricultural solutions, non-palm crops, and supply chain management systems. These are beyond the scope either deliberately, to keep the project relevant to its target users, or due to constraints such as time, dataset availability, and resource limitations as project does not attempt to replace professional agricultural systems but instead complements them by filling a gap for small-scale, entry-level palm owners.

8. Significant of The Study

We argue that this project will move the agriculture sector forward because it introduces an innovative approach to maintaining palm planting and the rich tradition behind it using advanced technologies such as image recognition and artificial intelligence. The project will provide customers with easy access to information related to palms and dates either via scanning, consultations, or articles in the mini wiki. The project will further support the digital transformation in agriculture in Saudi Arabia, as in spite of its significance, it has received limited attention for improvement.

8.1. Project Alignment with Saudi Vision 2030

The Nekhlawi project aligns with the strategic goals of Saudi Vision 2030 and principally the ones that focus on the agricultural, technology and culture industries [40]. The purpose of the project is to contribute to the aim of enhancing agricultural productivity and sustainability through the application of AI in the identification of diseases and monitoring the health of palm trees, which is why the use of modern technologies in the key spheres is also prioritized by 2030 Vision. Moreover, the use of AI in agriculture is also involved in the conservation of the Saudi Arabian agricultural heritage by computerizing the palm farming that constitutes a significant part of the cultural heritage and thus counted under the heritage preservation programs that form part of the Vision [41]. The project also provides digital tools that can transfer the knowledge of old experts to the young ones and thereby contributes to the attainment of the Vision 2030 of improving the status of the people and improving their quality of life by provision of non-discriminatory digital services. The small-scale palm owners are becoming more integrated into the national economy by opening themselves to the services of the expert consultants and guidance on agriculture and therefore, the rural communities are being considered as equals with regards to opportunities [40]. Finally, Nekhlawi conforms to the digital transformation of Vision and will utilize AI, machine learning, and mobile technologies to local agricultural issues, and focus on transforming the Kingdom economy into a more technologically oriented one and use smart solutions in every industry [40].

9. Professional and Ethical Implications

9.1. Professional Responsibilities:

- The project team should adhere to the accepted standards of software engineering (including IEEE and ACM) to make sure the system is accurate, reliable, and user-friendly.
- Disease detection is essential; hence, the AI models have to be trained and tested using quality datasets to eliminate fraud.
- Constant maintenance and updates are needed to keep the system sustainable and guarantee the quality of the services provided.

9.2. Security Responsibilities:

- Encryption and authentication are very essential to ensure the safety of the user data, such as palm tree images, consultations, and payment details.
- The system should be able to conduct regular security checks to avert hacking, breaches, or unauthorized use of sensitive data.
- It is extremely important to safeguard personal contact information of the professionals to prevent spamming and abuse.

9.3. Legal Responsibilities:

- Adherence to Saudi data protection legislation and foreign standards, e.g., GDPR, as necessary.
- Licensed or permissioned datasets are the only datasets that can be used without infringing intellectual property rights.
- Any financial transactions, such as paid consultations, have to be in accordance with local e-commerce laws and regulations.

9.4. Ethical Responsibilities:

- The implementation of AI-based diagnostics is to be applied in collaboration with human professionals in the field of agriculture but not instead of them.
- In the paid consultation model, it is important that the experts are treated with fairness in ensuring that their services are well appreciated.
- The project ought to embrace ethical data collection whereby the personal or agricultural data of the users should not be collected commercially without their knowledge.

- The project ought to embrace ethical data collection whereby the personal or agricultural data of the users should not be collected commercially without their knowledge.

- The issues of AI fairness must be considered in order not to misdiagnose due to the limitations of the data set. The dataset will be expected to represent the state of numerous palm trees to reduce bias and increase diagnostic accuracy.

Social Responsibilities:

- The venture ensures that palm farming is sustainable and that it helps in the maintenance of culture.
- The site can be made into educational content for the future generation, where the farming culture will be preserved in digital form.
- It promotes community building, especially the incompetence of the smallholder.
- Make the software user-friendly to people with low technical skills.

10. Project Organization

The report of this project is divided into seven major chapters; the specifics of this structure are as follows:

Chapter 1:

The introduction of the report is the first chapter, and it is a discussion on the background of the project, the problem statement, what the project seeks to achieve, its objectives, the scope, and its significance. Besides, the section also talks about the professional and ethical implications and provides a preview of the report structure. Furthermore, there is an elaborate overview of the available research and technological solutions in regard to the project. It discusses methodological techniques, such as conventional techniques, machine learning techniques, and deep learning techniques of palm-specific disease detection. Other important points regarding data handling, model evaluation, and applications are also covered in the chapter with the identification of the research gap that this project is meant to fill.

Chapter 2:

The title of the second chapter is the Software Project Management Plan (SPMP), which includes the scope of the project, assumptions, constraints, and deliverables. It details the management processes in detail, including the roles of the project, the requirements control, the schedule and budget management, the risk management, and quality control, and the supporting processes of documentation, configuration management, and validation.

Chapter 3:

The third one is the Requirements Specification, which is an indication that it produces the functional and non-functional requirements of the system. The chapter presents input-output description, requirement prioritization, and system analysis in accordance with the standards of IEEE SRS.

Chapter 4:

The fourth chapter is named "Design Specifications" and outlines system design and software and hardware specifications. It also has UML diagrams (use cases, classes, and sequences), database schema, system architecture, and UI mockups.

Chapter 5:

The fifth chapter is the conclusion, which is a summary of the study by giving the findings, contribution, and lessons learned. It puts light on the shortcomings of the project and offers future work suggestions.

CHAPTER 2: SOFTWARE PROJECT MANAGEMENT PLAN (SPMP)

1. Overview

1.1. Project Summary

1.1.1. Purpose

The purpose of this project is to preserve and safeguard the heritage of date palm cultivation in Saudi Arabia from the gradual decline caused by the younger generation's limited knowledge, neglect, and diseases. By providing an intelligent mobile application that offers direct diagnosis of palm tree diseases and pests, along with professional guidance and awareness for new palm owners, the project aims to ensure the continuity of agricultural and cultural practices.

This initiative represents a strategic effort to maintain date palms as both an agricultural and cultural legacy, contributing to the sustainability of palm cultivation and reducing unnecessary losses. Ultimately, the project seeks to protect the palm tree and its heritage from harmful practices carried out without proper knowledge.

1.1.2. Assumptions and Constraints

Assumptions

- App will provide scanning diagnoses, nutrition facts, consultations from experts and a mini wiki
- Computer vision and machine learning will be trained with the available datasets
- Users will have access to the app via smartphones and their smartphones' cameras
- Experts will be available to provide consultations either via chat, video calls, or in-person consultations
- User privacy and protection will be respected
- User must consent to sharing their data for app improvement purposes

Constraints

- Project must be developed within the timeline
- The collection of data must be aligned with the seasonal availability of dates
- Consultations cost must be affordable, considering all the factors such as distance
- The limited access to datasets may lead to slowing down the accuracy
- Data protection laws must be followed

1.1.3. Project Deliverables

Table 3: Project Deliverables

Deliverable	Description	Delivery Date	Delivery Location	Quantity/Format	Special Instructions
Proposal discussion	Online team meetings with supervisor to discuss initial project idea, scope, and objectives.	4/9/2025	Zoom	4 meetings	Take notes and make sure to record the meeting for further future notes.
Collecting data and defining the domain of the problem (Problem statement)	outlining the research domain, key challenges, and finalized problem statement.	18/09/2025	Zoom and OneDrive	1 pptx file	Collect data from similar project ideas.
Project proposal document	Formal project proposal describing objectives, expected outcomes, challenges, gaps.	20/9/2025	Zoom and OneDrive	1 pptx file	none
Literature review	A collection of academic and technical papers related to palm tree diagnosis, care, and used technologies.	24/09/2025	OneDrive	25 pdf files	5 LR for each team member.
Project management plan (SPMP)	A structured project plan covering scope, schedule, resources, risks, and management approach.	09/10/2025	Blackboard	1 docx file	Use IEEE Std 1058-1998 Guidance.

Deliverable	Description	Delivery Date	Delivery Location	Quantity/Format	Special Instructions
Midterm report	A progress report summarizing completed work, challenges faced, and next steps toward project completion.	16/10/2025	Blackboard	1 docx file	Follow college's report template.
System requirements specification (SRS)	Detailed functional and non-functional requirements of the app	30/10/2025	Blackboard	1 docx file	Use IEEE SRS format.
Initial design	Preliminary design document including system architecture, database schema, and user interface mockups.	13/11/2025	Blackboard	1 docx file	Include UML diagrams (use case, class, sequence) + ensure UI mockups are attached.
Final design	Completed system design document including refined architecture, finalized database schema, and UI/UX layouts.	4/12/2025	Blackboard	1 docx file	Include refined database schema + complete system architecture diagram.
Final report	Comprehensive document summarizing the entire project, including research background, methodology, system design, implementation, testing results, and conclusions.	19/12/2025	Blackboard	1 docx file	Delivered in a software copy + include references in IEEE style.

Deliverable	Description	Delivery Date	Delivery Location	Quantity/Format	Special Instructions
Final oral presentation	Live presentation to showcase the project outcomes, demonstrate the app, and answer evaluation committee questions.	08/01/2026	In-class / Zoom (if online) & Blackboard	1 pptx file	all team members must present + rehearse demo in advance.

1.1.4. Master Schedule and Budget Summary

Master Schedule

https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

TASK	ASSIGNED TO	PROGRESS	START	END
Initiation				
Define project idea	All members	100%	8/25/25	8/26/25
Identify stakeholders	All members	90%	8/26/25	8/29/25
Feasibility study	All members	90%	8/29/25	9/1/25
Define project scope	All members	100%	9/1/25	9/2/25
Obtian approval	Zainab Hassan	100%	9/2/25	9/5/25
Planning and design				
Divide tasks	Zainab Hassan	55%	9/5/25	9/8/25
Idea generation and Justification	All members	100%	9/8/25	9/11/25
Project Definition	All members	100%	9/11/25	9/14/25
Status report 1	Doja Alnemer and Zainab Al-rashed	100%	9/14/25	9/17/25
SPMP (Section 1 + Section 2)	All members	100%	9/17/25	9/22/25
Midterm report (Chapter 1: Introduction)	All members	100%	9/17/25	9/22/25
Status report 2	All members	100%	9/22/25	9/27/25
SPMP(Section 3 + Section 4)	All members	85%	9/27/25	10/6/25
Survey qastions and publishing	All members	90%	9/27/25	10/6/25
Contact with organizitions	Doja Alnemer	0%	9/27/25	10/6/25
SPMP(Section 5)	All members	35%	10/6/25	10/3/25
Research papers reading	All members	35%	10/6/25	10/4/25
Liteurer review	All members	45%	10/6/25	10/4/25
SPMP (Section 1 + Section 2)	All members	100%	9/17/25	9/22/25
Midterm report (Chapter 1: Introduction)	All members	100%	9/17/25	9/22/25
Status report 2	All members	100%	9/22/25	9/27/25
SPMP (Section 3 + Section 4)	All members	85%	9/27/25	10/6/25
Survey qastions and publishing	All members	90%	9/27/25	10/6/25
Contact with organizitions	Doja Alnemer	0%	9/27/25	10/6/25
SPMP (Section 5)	All members	35%	10/6/25	10/3/25
Research papers reading	All members	35%	10/6/25	10/4/25
Liteurer review	All members	45%	10/6/25	10/4/25
SPMP (Section 6 + Section 7)	All members	0%	10/4/25	10/10/25
Status report 3	Zainab Albadi and Hebah Altheyab	80%	10/10/25	10/13/25
Project logo and name	All members	0%	10/10/25	10/11/25
SPMP (Section 8)	All members	0%	10/11/25	10/16/25
SPMP (Appendices)	All members	0%	10/16/25	10/16/25
SPMP review and midterm sumbission	All members	0%	10/16/25	10/19/25
Midterm report sumbision	All members	0%	10/16/25	10/19/25
Analyzing phase (System requirements) (SRS)	All members	0%	10/19/25	10/26/25
Analyzing phase (System requirements) (SRS)	All members	0%	10/26/25	11/2/25
Status report 4	All members	0%	11/2/25	11/3/25
SRS review (Chapter 4)	All members	0%	11/2/25	11/3/25
Initial design (SDS)	All members	0%	11/2/25	11/7/25
Initial design (SDS)	All members	0%	11/7/25	11/14/25
Status report 5	All members	0%	11/13/25	11/14/25
SDS review (Chapter 5)	All members	0%	11/13/25	11/14/25
Final design	All members	0%	11/14/25	11/30/25
Final design	All members	0%	11/30/25	12/7/25
Final Report	All members	0%	11/30/25	12/7/25
Presentation file	All members	0%	12/7/25	12/10/25
Presentation revie	All members	0%	12/10/25	12/11/25
Oral presention	All members	0%	12/16/25	12/26/25
Execution				
Database construstion (user part)	Zainab Albadi and Hebah Altheyab	0%	1/2/26	1/5/26
Dataset prepration	Zainab Albadi	0%	1/5/26	1/8/26
Prepare the required software tools	All members	0%	1/8/26	1/7/26
UI/Ux design	All members	0%	1/7/26	1/14/26
AI model prepration and integration	All members	0%	1/7/26	1/14/26
App deployment	All members	0%	1/14/26	1/24/26
Presention prepration	All members	0%	1/24/26	1/29/26
Evaluation				
Verification (Unit test and Review document)	Zainab Albadi and Doja Alnemer	0%	1/14/26	1/24/26
Integration Testing	Hebah Altheyab	0%	1/21/26	1/28/26
Validation(Uasablilty test)	Zainab Alrashed	0%	1/28/26	2/4/26
Quality Oversight	Donia Alfarhan	0%	2/4/26	2/11/26
Final report review	All members	0%	2/11/26	2/14/26
Prestention Document reviw	All members	0%	2/11/26	2/14/26
Close out				
Gradution project exhibition	All members	0%	2/17/26	2/18/26
Graduation project publish	All members	0%	2/18/26	2/19/26
Final report submision	All members	0%	2/19/26	2/26/26
Final oral presentation	All members	0%	2/26/26	3/5/26

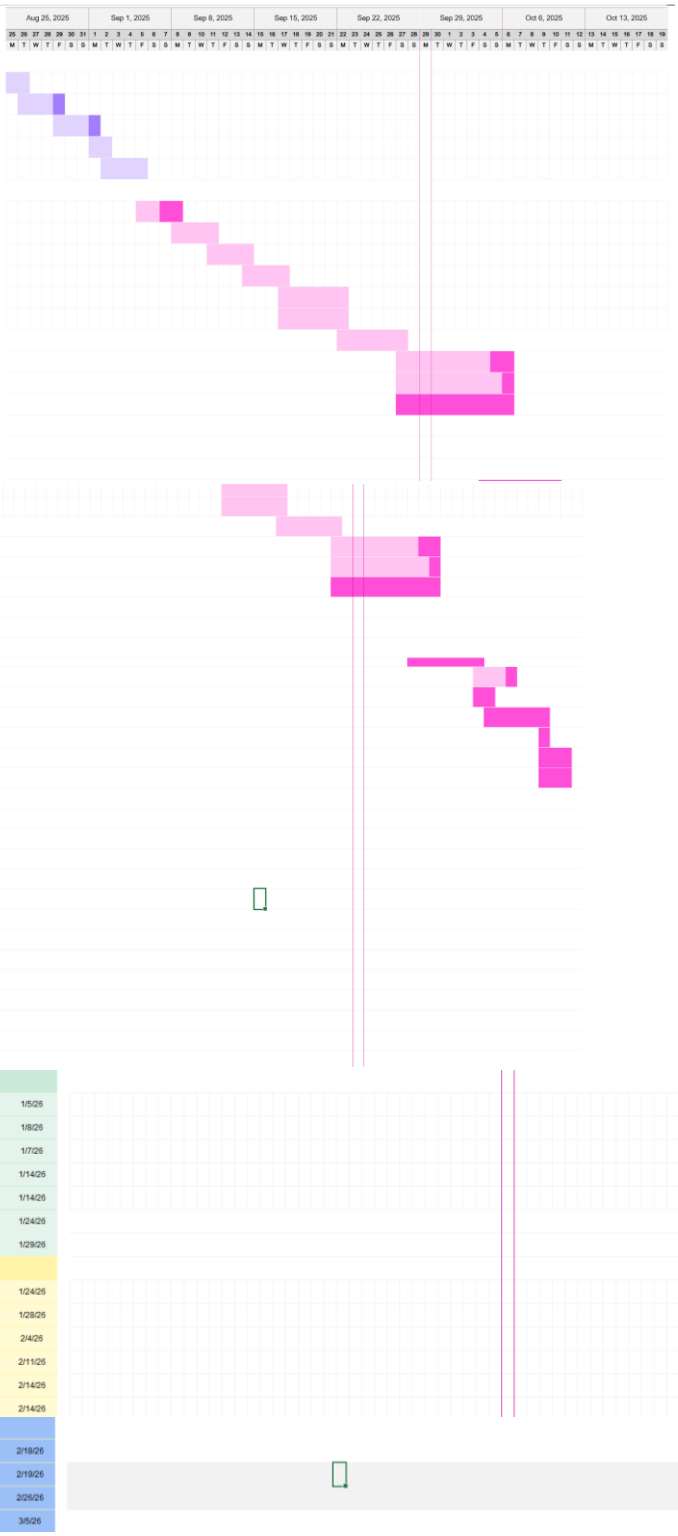

Figure 2: Master Schedule

Budget Summary

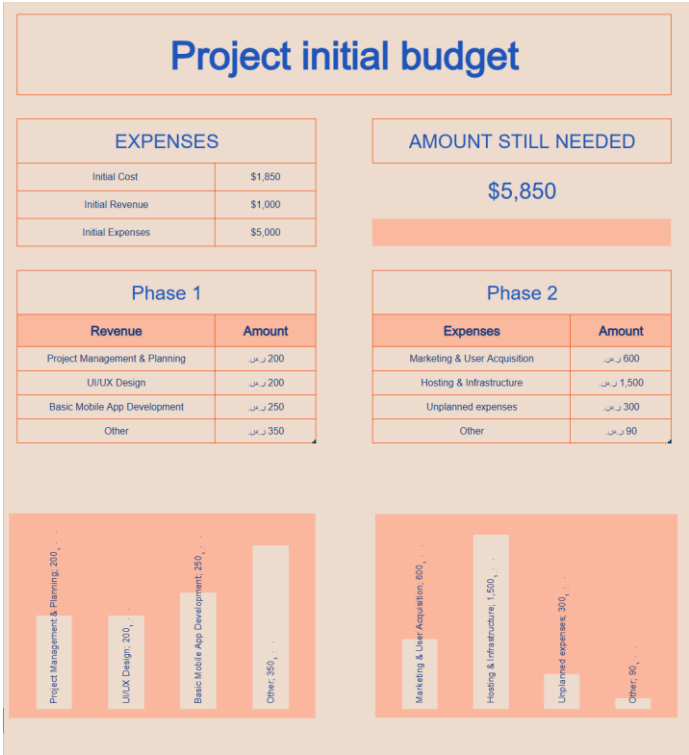

Figure 3: Project initial budget

1.2. Evolution of the Plan

The project management plan (PMP) is a dynamic document that will change during the project lifecycle. Updates will take place based on the project processes, risk event inputs, stakeholders feedback, or any changes in schedule, resources, or scope.

Scheduled updates:

On the completion of major milestones, formal updates will take form, including:

- Project proposal document
- Midterm report
- Final design
- Final report
- Final oral presentation

Unscheduled updates:

Updates might be made whenever needed to reflect any changes in requirements, unexpected issues, or risks. Updates can be proposed by any member of the team, then evaluated and approved by the project manager before being implemented.

Dissemination of updates:

- Any accruing updates will be shared with the team members and supervisor via OneDrive and Blackboard.
- Update notifications will be sent to all team members within the WhatsApp group and communicated during weekly meetings.

Configuration Management:

- The initial draft version of the document will be preserved under configuration management in OneDrive.
- There will be a version number for each revision (e.g., v1.0, v1.1, v2.0).
- A Record of Changes table will be maintained for the text, indicating each change, the author, date, and description, as well as a reference to the change request where applicable.
- Version control is the project manager's responsibility, and only the latest approved version should be distributed.
- By using this structured approach, the PMP will be capable of adjusting to evolving needs while maintaining consistency, accuracy, and alignment with project objectives.

1.3. Document Structure

This plan is organized as follows:

a. Section 1, Project Overview. This section provides an overview of the scope and objectives of the project, the project's assumptions and constraints, reference to the project deliverables, schedule and budget, and a description of the evolution of the plan.

b. Section 2, References. This section provides a list of all documents, policies, templates, processes, and other sources of information referenced in the plan.

c. Section 3, Definitions. This section contains the abbreviations and acronyms required to properly understand this planning document.

d. Section 4, Project Organization. This section identifies interfaces to organizational entities external to the project, the project's internal organizational structure, and defines roles and responsibilities for the project.

e. Section 5, Management Process. This section describes the planning, measurement, tracking, reporting, risk control mechanisms needed to provide management control over the technical processes and product quality, and appropriate project initiation and closeout procedures.

f. Section 6, Technical Process. This section describes the technical solution in terms of a process model and implementation methods, tools, and techniques to be used to develop the various work products, plans for establishing and maintaining the project infrastructure, and the product acceptance.

g. Section 7, Supporting Processes. This section describes processes that are employed to facilitate and control the technical processes and the state of the product. These include, but are not limited to, configuration management, verification and validation, documentation, quality assurance, reviews and audits, problem resolution, and contractor management, and methods to ensure continuous process improvement.

h. Section 8, Additional Plans. This section addresses the logistic support strategy to be applied to increase the system's operational effectiveness.

2. Definitions

Table 4: Table of Definitions

Abbreviation	Definition
UML	Unified Modeling Language.
IEEE	Institute of Electrical and Electronics Engineers.
AI	Artificial Intelligence.
ML	Machine Learning.
ERP	Enterprise Resource Planning.
APIs	Application Programming Interfaces.
UI	User interface.
SPMP	Software Project Management Plan.

Abbreviation	Definition
SRS	System Requirements Specification.

3. Project Organization

3.1. External Interfaces

There are a few external entities that the project will interface with and that will in some way or other assist or influence the project. Such interfaces will be declarative of the organizational roles and limits between the project team members and the external stakeholders:

Parent organization:

The team behind the palm tree care mobile application design, creation, and maintenance is CIS major students of Imam Abdulrahman Bin Faisal University (College of Computer Sciences and IT).

Acquiring Organization:

The main end users are small-scale owners of palms, including households, and people around Saudi Arabia who will access the mobile application for the diagnosis of palm disease, seasonal care tracking, and professional consultation.

Subcontracted Organizations:

- Loud Service Providers: Store image recognition, consultation, and data storage backend infrastructure.
- Agricultural Experts/Consultants: are given expert consultation services and domain knowledge and act as external specialists to validate palm diagnoses.

Other Organizational Entities:

- Governmental Agricultural Authorities: Can be contacted with the purpose of connecting with the national projects that help to digitize agriculture and preserve palm heritage.
- Community Farmers and Senior Palm Growers: They work as sources of knowledge whose traditional experience is added to the knowledge base of the app.

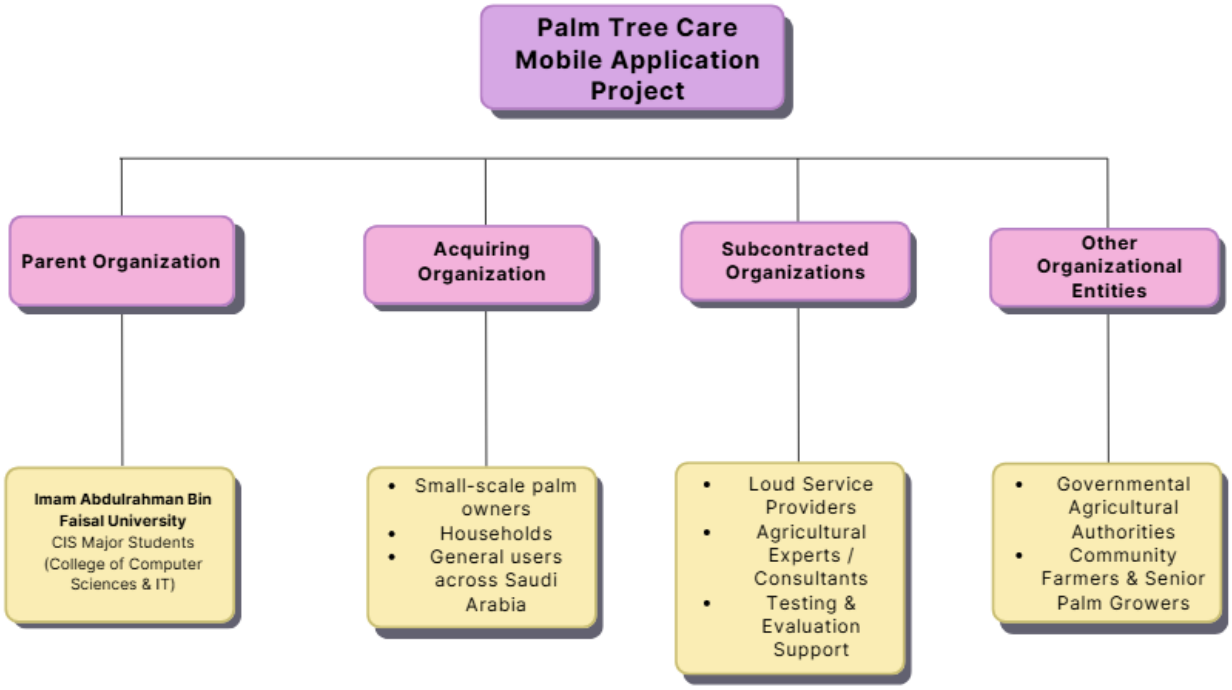

Figure 4: External interface organizational chart

3.2. Internal Structure

This project will be supervised by Mrs. Albandri Alamer will ensure its timely and successful completion. This project will be developed and executed by Zainab Al-badi, Hebah Altheyab, Zainab Al-rashed, Doja Alnemer, Donia Alfarhan. Each member is assigned a role that aligns their strength, skills and experience, thereby ensuring the quality in every stage of the project.

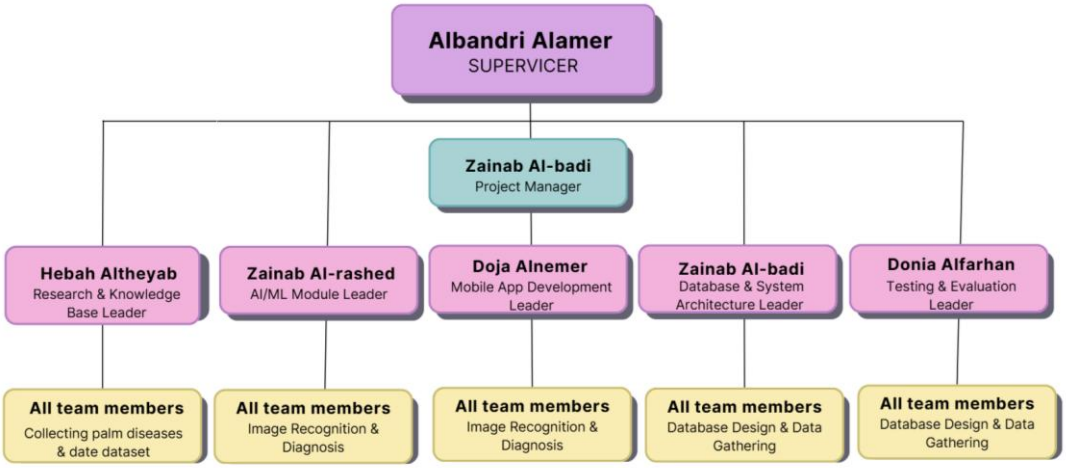

Figure 5: Organizational chart

3.2.1. The Project Manager

The Project Manager, Zainab AL-Badi has a role in guiding the team toward the successful completion of the project. This role involves leadership, decision-making, and ensuring that all team members are involved in the project and that project objectives are achieved. The Project Manager also serves as a bridge between the team and the supervisor, maintaining smooth communication and accountability throughout the project lifecycle.

3.2.1.1. Scope of Responsibility

The Project Manager responsibility is to oversee the overall progress of the project and ensure that the objectives are met within the defined scope, schedule, and quality standards. This includes monitoring task completion, tracking milestones, managing risks, and ensuring compliance with project requirements. The Project Manager must also ensure that the team remains aligned with the project goals, while maintaining effective documentation and reporting throughout the project lifecycle.

3.2.1.2. Internal Responsibilities

Internally, the Project Manager is responsible for coordinating team efforts, distributing tasks fairly, and ensuring that each member understands their role. The Project Manager facilitates regular meetings, provides guidance, and addresses any conflicts or misunderstandings among the members. Additionally, the Project Manager monitors

the quality of each contribution and ensures that all team activities are synchronized to meet the internal deadlines.

3.2.1.3. External Responsibilities

Externally, the Project Manager serves as the official representative of the team to the supervisor and any other external stakeholders. Responsibilities include submitting deliverables on time, providing progress updates, addressing supervisor feedback, and ensuring that the project complies with academic and ethical standards. The Project Manager also communicates any challenges or risks that may affect deliverables and seeks guidance from the supervisor when needed.

3.3. Project Roles and Responsibilities

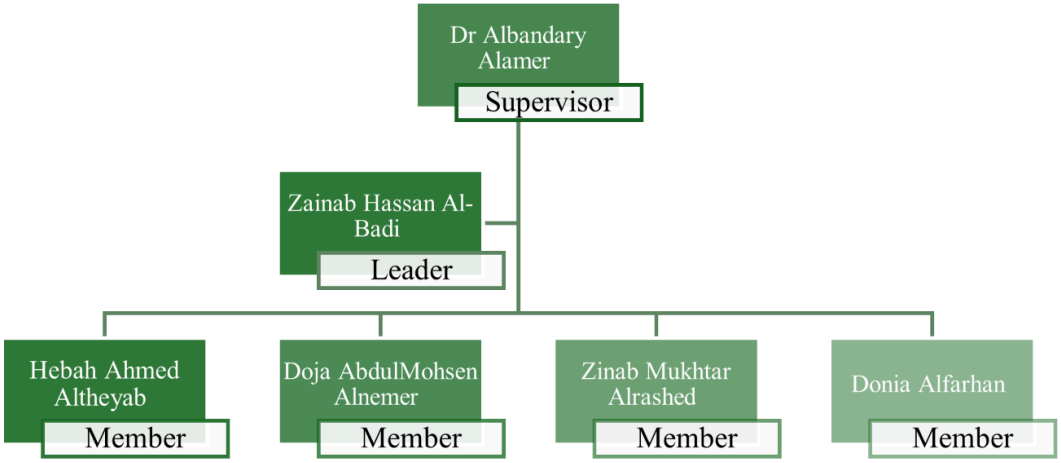

Figure 6: Project Roles and Responsibilities

Table 5: Responsibility matrix

Task / Member	Dr Albandary Alamer	Zainab Albadi	Hebah altheyab	Doja Alnemer	Zainab Alrashed	Donia Alfarhan
Project Management	C	R / A	C	I	I	I
System Analysis	I	R / A	A	R / A	A	A
System Design	C	R / A	R	A	A	R / A
Interface Design	I	R	R / A	R / A	R	R
Programming	I	R	R	R / A	R / A	R
Testing and Debugging	I	R	R	R	R	R / A
Documentation	I	R	R / A	R	R	R / A

Key

R: Responsible (the member that performs the activity)

A: Accountable (The member that's accountable for the activity's success)

C: Consulted (The member that provide input or expertise during the activity)

I: Informed (The member that needs to be kept informed about progress and decisions)

4. Management Process

This section provides an overview of how Nekhlawi project will be managed using start up plan, work plan, resource and schedule allocation, project controls and risk management.

4.1. Start-up

This section is part of the project management plan to specifies the estimation of cost and persona of the Nekhlawi project.

4.1.1. Estimation

Estimation will be conducted from the team's previous knowledge and experience working on other projects using a variety of tools and methods. The project implementation duration will be around 3 months, and the evaluation will be done periodically during team meetings. The project will be divided into smaller tasks, and teams will put equal effort into achieving these tasks, using 3 hours a day to do so. Considering the low budget of the team, free or affordable tools will be used, as shown in the table.

Table 6: Tools to be used and their cost

Tool	Cost
Microsoft Office Package	Free
Zoom	Free
Visual Studio	Free
Draw. Io	Free
GitHub	Free
MySQL	Free

4.1.2. Staffing

The team is made up of 5 members with multiple skills that are required throughout the different phases of project implementation. The table below specifies the number of members required for each role and level of skill, moreover, the phase needed for it.

Table 7: Staff roles, skills and phases

Role/Skill required	Skill level	No. members	Phases
Project manager	High	1	All phases
Full-Stack developer	Medium	3	All phases
Front-end developer	High	1	• Software requirement specification • Software design specification • Implementation • Testing
Back-end developer	High	1	• Software requirement specification • Software design specification • Implementation • Testing
Database developer	High	2	• Software requirement specification • Software design specification • Implementation
AI/ML Specialist	High	3	• Software requirement specification • Software design specification • Implementation • Testing
Communication skills	High	5	All phases

4.1.3. Resource Acquisition

Table 8: Resource Acquisition

Resource	Description	Responsibility
Software	All software will be downloaded by team members from their original websites: • Microsoft Office • Zoom • Visual Studio • Draw.io • GitHub • MySQL	All members
Hardware	Team members will use their own devices to work on the project.	All members
Data	Data will be collected from trusted sources in order to train the AI model and data quality will be evaluated by project manager and AI/ML Specialists.	• Project manager • AI/ML Specialists
Human resource	Team members' knowledge, experience and skills will be used and developed throughout different phases of the project.	All members
Services and outside resources	Communicating with outsiders to gain access to different data and services.	Project manager

4.1.4. Staff Training

Team members will be further trained to improve their skills and use the required tools to implement the project. The training will focus on AI/ML and mobile application development throughout, taking courses online and consultations from expert doctors at the university. Moreover, the team will boost their knowledge in the agricultural domain by searching for information and reaching out to experts for their opinions and knowledge.

4.2. Work Planning

4.2.1. Work Activities

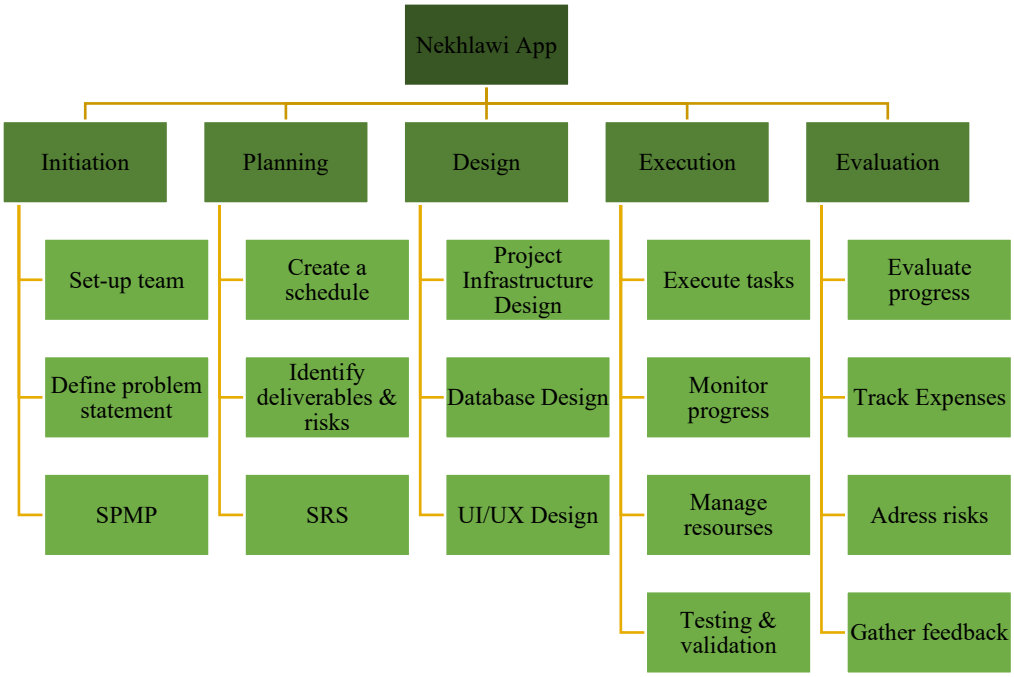

Figure 7: Work activities

4.2.2. Schedule Allocation

Schedule

TASK	ASSIGNED TO	Resources		Duration	START	END
Initiation		Software	Hardware			
Define project idea	All members	There is no used recourses		1 day	8/25/25	8/26/25
Identify stakeholders	All members			3 days	8/26/25	8/29/25
Feasibility study	All members			3 days	8/29/25	9/1/25
Define project scope	All members			1 day	9/1/25	9/2/25
Obtian approval	Zainab Hassan			3 days	9/2/25	9/5/25
Planning and design						
Divide tasks	Zainab Hassan	Excel and WhatsApp	Personal PC	3 days	9/5/25	9/8/25
Idea generation and Justification	All members	There is no used recourses	There is no used recourses	3 days	9/8/25	9/11/25
Project Definition	All members	There is no used recourses	There is no used recourses	3 days	9/11/25	9/14/25
Status report 1	Doja Alnemer and Zainab Al-rashed	Microsoft word and OneDrive	PC	3 days	9/14/25	9/17/25
SPMP (Section 1 + Section 2)	All members	Microsoft word and OneDrive	PC	5 days	9/17/25	9/22/25
Midterm report (Chapter 1: Introduction)	All members	Microsoft word and AI models and OneDrive	PC	5 days	9/17/25	9/22/25
Status report 2	All members	Microsoft word and AI models and OneDrive	PC	5 days	9/22/25	9/27/25
SPMP (Section 3 + Section 4)	All members	Microsoft word and AI models and OneDrive	PC	9 days	9/27/25	10/6/25
urvey questions and publishing	All members	Google forms	PC	9 days	9/27/25	10/6/25
Contact with organiztions	Doja Alnemer	Microsoft outlook		9 days	9/27/25	10/6/25
SPMP (Section 5)	All members	Microsoft word and AI models and OneDrive	PC	9 days	9/27/25	10/6/25
Research papers reading	All members	AI models and OneDrive	PC	9 days	9/27/25	10/6/25
Liteurer review	All members	AI models and OneDrive	PC	9 days	9/27/25	10/6/25
SPMP (Section 6 + Section 7)	All members	Microsoft word and AI models and OneDrive	PC	6 days	10/4/25	10/10/25
Status report 3	Zainab Albadi and Hebah Altheyab	Microsoft word and AI	PC	3 days	10/10/25	10/13/25

		models and Onedrive				
Project logo and name	All members	Canva	PC	1 day	10/10/25	10/11/25
SPMP (Section 8)	All members	Microsoft word and AI models and Onedrive	PC	5 days	10/11/25	10/16/25
SPMP (Appendices)	All members	Microsoft word and AI models and OneDrive	PC	24 hours	10/16/25	10/16/25
SPMP review and midterm sumbision	All members	Microsoft word and AI models and OneDrive	PC	3 days	10/16/25	10/19/25
Midterm report sumbision	All members	Blackboard	PC	3 days	10/16/25	10/19/25
Analyzing phase (System requirements) (SRS)	All members	Draw.io and Microsoft word and Onedive	PC	7 days	10/19/25	10/26/25
Analyzing phase (System requirements) (SRS)	All members	Draw.io and Microsoft word and Onedive	PC	7 days	10/26/25	11/2/25
Status report 4	All members	Microsoft word and AI models and Onedrive	PC	1 day	11/2/25	11/3/25
SRS review (Chapter 4)	All members	Microsoft word and AI models and Onedrive	PC	1 day	11/2/25	11/3/25
Initial design (SDS)	All members	Blasmiq and Figma	PC	5 days	11/2/25	11/7/25
Initial design (SDS)	All members	Blasmiq and Figma	PC	7 days	11/7/25	11/14/25
Status report 5	All members	Microsoft word and AI models and Onedrive	PC	1 day	11/13/25	11/14/25
SDS review (Chapter 5)	All members	Microsoft word and AI models and OneDrive	PC	1 day	11/13/25	11/14/25
Final design	All members		PC	6 days	11/14/25	11/20/25
Final design	All members		PC	7 days	11/30/25	12/7/25
Final Report	All members		PC	7 days	11/30/25	12/7/25
Presentation file	All members	Microsoft pptx and AI models and OneDrive	PC	3 days	12/7/25	12/10/25
Presentation review	All members	Microsoft pptx and AI models and OneDrive	PC	1 day	12/10/25	12/11/25
Oral presentation	All members	Zoom or Teams	PC	2.5 week	12/16/25	12/26/25

Execution						
Database construction (user part)	Zainab Albadi and Hebah Altheyab	MySQL	PC	3 days	1/2/26	1/5/26
Dataset preparation	Zainab Albadi	MySQL	PC	1 day	1/5/26	1/8/26
Prepare the required software tools	All members	PC	PC	1 day	1/8/26	1/7/26
UI/Ux design	All members		PC	7 days	1/7/26	1/14/26
AI model preparation and integration	All members	Axure cloud	PC	7 days	1/7/26	1/14/26
App deployment	All members	Mobile phone	PC	10 days	1/14/26	1/24/26
Presentation preparation	All members	Microsoft pptx and AI models and OneDrive	PC	5 days	1/24/26	1/29/26
Evaluation			PC			
Verification (Unit test and Review document)	Zainab Albadi and Doja Alnemer		PC	10 days	1/14/26	1/24/26
Integration Testing	Hebah Altheyab		PC	7 days	1/21/26	1/28/26
Validation (Usability test)	Zainab Alrashed		PC	7 days	1/28/26	2/4/26
Quality Oversight	Donia Alfarhan		PC	3 days	2/4/26	2/11/26
Final report review	All members		PC	3 days	2/11/26	2/14/26
Presentation Document review	All members	Microsoft pptx and AI models and OneDrive	PC	3 days	2/11/26	2/14/26
Close out						
Gradution project exhibition	All members		-	1 day	2/17/26	2/18/26
Graduation project publish	All members		PC	1 day	2/18/26	2/19/26
Final report submission	All members	Blackboard	PC	7 days	2/19/26	2/26/26
Final oral presentation	All members	Zoom or Teams	PC	7 days	2/26/26	3/5/26

Figure 8: Schedule Allocation

4.2.3. Resource Allocation

Table 9: Resource Allocation

<table>
<tr><th>Activity</th><th>Material Resources</th><th>Human Resources</th></tr>
<tr><td>Problem statement</td><td rowspan="5">• Personal Computers
• OneDrive
• MS Word
• MS Excel
• Draw.io

• Figma
• MySQL Workbench
• Wireframes</td><td rowspan="6">• Team members
• Team Leader
• Team Supervisor</td></tr>
<tr><td>Project plan (SPMP)</td></tr>
<tr><td>Midterm report</td></tr>
<tr><td>System Requirements specification (SRS)</td></tr>
<tr><td>Software Initial design</td></tr>
<tr><td>Project documentation</td><td>• MS Word
• MS Excel
• OneDrive</td></tr>
<tr><td>Final report & Oral presentation</td><td>• MS Word
• MS Power point</td><td>• Team members
• Team Leader
• Team Supervisor
• Evaluators</td></tr>
</table>

4.2.4. Budget Allocation

Table 10: Budget Allocation

Software	Software cost
Microsoft Office Package	Free (by using university email)
WhatsApp	Free
Zoom	Free
Figma and Balsamiq (UX/UI)	200 SAR
Draw.io and BPMN	Free
PC	Free

4.3. Project Controls

This section provides the techniques employed in the management and monitoring of the project. These controls are used to make sure that the project is within schedule, scope, and of high quality.

4.3.1. Requirements Control

The requirement control plan will be adopted to ensure that all non-functional and functional requirements of the Palm Diagnosis and Awareness App are addressed properly. Once all the requirements are listed, a meeting shall be arranged with the supervisor, and this shall be to ensure that nothing has been left out. Any changes approved in the course of the development will be recorded and kept in the requirement control log at OneDrive. Once the development is done then the team will check every requirement with the aim of making sure that it is implemented properly in the system.

4.3.2. Schedule Control

A Gantt chart and Google Calendar will be used to control the project schedule and describe all stages and milestones. All the members of the team should ensure that they deliver by the expected deadlines. Status reports will be weekly, summarizing the tasks completed, ongoing, and delayed. The changes of schedules will be reduced to a minimum level to avoid delays; however, in case of necessity, the supervisor will approve the changes, and they will be reflected in the revised Gantt chart.

4.3.2.1. Schedule Tracking

Deadlines will be monitored using Google Calendar. The tasks will be arranged as events with reminders, and progress updates will be frequently taken.

4.3.2.2. Schedule Performance Reports

The members will have to accomplish given tasks based on their skills and schedules. The team leader will check on progress twice a week and provide assistance in case of any obstacle caused by a member.

4.3.2.3. Schedule Reviews

A review meeting will be conducted at the end of each milestone to compare the scheduled progress and actual progress. Such reviews make sure that they identify delays or work imbalances early.

4.3.2.4. Progress Variance Monitoring

The team will frequently monitor the variances in the progress of the planned and actual schedules. The schedule will include buffer time that will help to address unforeseen factors and avoid the feeling of time pressure at the end of the day.

4.3.2.5. Progress Variance Resolution

In case there are schedule delays, the tasks will be reallocated, or priorities reallocated to stabilize the project back on its course.

4.3.2.6. Follow-up on Corrective Action

Once the corrective measures are taken, the next meeting will be a review of whether the corrective measures will effectively address the problem next week.

4.3.3. Budget Control

Nekhlawi Application estimated budget is around 6000 SR. This includes all phases of development, which will be organized by the team leader. In case the cost is more than this, the team will seek further assistance from the university. To keep costs under control, the team will focus on freely available or student-licensed tools, i.e., Balsamiq and Figma.

4.3.3.1. Cost Management

All the expenses will be noted on a common Excel sheet in OneDrive and checked after every week.

4.3.3.2. Methods to Ensure Cost Adherence

The team will choose the open or free options to ensure that it does not spend beyond its budget.

4.3.3.3. Cost Control

Any spending must have the consent of the supervisor.

4.3.3.4. Contractor Cost Control

In case the external experts are brought in, the supervisor will authorize any associated expenditure prior to involvement.

4.3.3.5. Cost Variance Measurement

Variations of planned and actual costs will be recorded in the budget tracker.

4.3.3.6. Cost Variance Corrective Action

In case overspending is identified, unnecessary purchases will be delayed or substituted by the free alternatives.

4.3.4. Quality Control

Quality will be ensured by holding frequent meetings and in-house reviews. All the deliverables will be verified to be complete, accurate, and properly formatted before they are submitted. In case of any problems detected, the member concerned has to rectify them immediately. Files will be shared, and updates will be effectively reviewed via WhatsApp and OneDrive.

4.3.5. Project Reporting and Communication

Constant reporting and communication are the key to the success of the project. The members will have clear and consistent communication with the supervisor.

4.3.5.1. Electronic Media

Primary tools for communication and collaboration are Zoom, WhatsApp, OneDrive, email, and Google Calendar. Coding and design will also be done using development tools like VS Code, Figma, and MySQL Workbench.

4.3.5.2. Meetings

It will also have weekly meetings with the supervisor to discuss progress, challenges, and the next steps. Internal meetings will also be established to coordinate team members as required.

4.3.5.3. Information Repository

All the documents and deliverables will be saved in a common OneDrive folder with a master Word document to collaboratively write and provide version control.

4.3.5.4. Reviews

All the members will check any work by looking through it to make sure that it is correct and complete before handing the supervisor over the deliverable.

4.3.5.5. Status Reporting.

Meetings will be summarized in status reports, and assigned tasks will be completed in these reports. The reports will be presented on the Blackboard or directly to the supervisor.

4.4. Risk Management

Table 11: Project risk management plan

No.	Risk	Description	Impact	Response	Risk Level
1	Schedule Risks	Mistakes in estimating the time required to complete the task. It takes longer than expected.	Not meeting deadlines.	Apply Agile, short sprints, weekly follow-ups and cooperate to complete the task quickly.	Medium
2	Budget Issues	Limited financial resources.	Inability to obtain paid tools and services.	Use open-source tools and free tools provided by the university.	Medium
3	Scope Creep	Changes or ambiguity in project requirements.	The program may fail to meet expected deliverables.	Maintain clear documentation, review requirements with supervisor regularly.	low
4	Integration Risks	Problems connecting data, AI model, and the program in the cloud.	The program is unable to access the AI model and complete the task.	Early testing of integration, prepare backup cloud provider, document APIs.	High

No.	Risk	Description	Impact	Response	Risk Level
5	Dataset Collection Risk	Difficulty collecting datasets or missing samples.	Incomplete or low-quality training data affects accuracy.	Collaborate with external palm agricultural entities, apply data augmentation, and use public datasets.	High
6	Technological Risks	Software bugs or failure during development.	System crashes	Continuous testing and debugging sessions.	High
7	Skill Gaps	Lack of expertise in AI, cloud, or development tools.	Reduced quality of work, delays in progress.	Conduct training, seek external support, and use ready-made templates that are easy to understand and use.	Medium
8	Data Loss Risks	Loss of project files due to error or failure.	Losing critical project progress and documentation.	Use GitHub and maintain local cloud backup.	High
9	Low Survey Response Risk	Not enough survey responses from potential users.	The application does not meet the users' needs.	Disseminate the survey more widely and run advertising campaigns to reach the largest number of people.	Low

5. Technical Process

5.1. Process Model

In this project, a waterfall development methodology will be followed, as it is carefully planned before starting implementation with specified requirements, and the work of each phase must be done before moving on to the next. This

methodology helps the team focus on the task and be aware of the tasks that must be done throughout the five stages shown below in the figure.

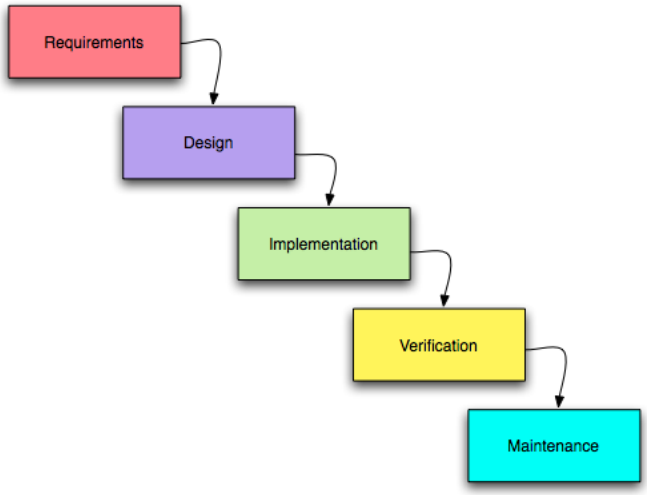

Figure 9: Waterfall Model Stages

5.2. Methods, Tools and Techniques

The methods used in this project are the waterfall development and the Gantt charts to keep track of the work.

Tools that will help accomplish the project implementation are:

1. Microsoft office
2. Draw.io
3. Zoom
4. Diagrams
5. Google and Google Scholar
6. AI suitable programming languages
7. MySQL Workbench

However, the technique that we will follow is the IEEE standards.

5.3. Project Infrastructure

The Nekhlawi project's development and testing environment consists of the team's personal devices running Windows or macOS, each equipped with at least an i5 processor and 8 GB of RAM or higher.

The main development tools include Visual Studio, Android Studio, Flutter, Figma, and Amazon Web Services (AWS) for cloud hosting and deployment.

The team primarily works remotely using its own facilities. To support this setup, all project data and documentation are stored on a shared OneDrive, while the source code is maintained in a GitHub/Git repository to ensure version control, compatibility, and collaborative development.

When necessary, the team meets at Imam Abdulrahman Bin Faisal University facilities for project discussions, progress reviews, and supervisor meetings.

5.4. Product Acceptance

The Nekhlawi project deliverables will be formally accepted by the academic supervisor and the evaluation committee at Imam Abdulrahman Bin Faisal University.

The acceptance process will ensure that the developed system and its documentation meet all functional and non-functional requirements defined in the SRS.

Acceptance Methods:

- Demonstration: The team will conduct a live demonstration of the mobile application, showing its main functions including palm disease diagnosis, date type identification, and expert consultation interface.
- Testing: The system will undergo usability and accuracy testing to verify that its performance and functionality align with the stated requirements.
- Inspection: All documentation (SPMP, SRS, SDS, and user manual) will be reviewed for completeness, clarity, and compliance with IEEE standards.

Acceptance Criteria:

- The application runs smoothly on Android devices without major defects.
- All primary features described in the SRS are implemented and functional.
- Diagnosis accuracy reaches an acceptable level (≥ 90%).
- The user interface is intuitive and accessible.
- All required documents are delivered on time and in the approved format.

Approval and Sign-off:

Final acceptance will occur after successful presentation and evaluation by the supervisor and project committee. A signed approval document will confirm that the deliverables meet the acceptance criteria.

6. Supporting Processes

6.1. Configuration Management

Configuration management is supposed to monitor and regulate all of the changes that have been applied to the important project papers and deliverables. This is an important process that will be used to ensure consistency and to ensure that the objectives of the project are maintained throughout its lifecycle. All modifications, be they in accordance with requirements, design requirements, or code, should be checked and validated by the supervisor before any implementation can take place. The implementation of this rigid process prevents any element of the project from working independently in an imperfect manner and keeps the team working effectively at all times, keeping the standard high throughout the project.

6.2. Independent Verification and Validation

This section outlines the Verification and Validation plan for accurate development of Nekhlawi App.

Verification Activities

- Code Review: Ensuring that the source code adheres to the best practices and standards in software engineering.
- Design Validation: Verifying that UML diagrams, database schemas, and UI mockups align perfectly with the written requirements.
- Algorithm Evaluation: Confirming that the AI and image recognition algorithms designed to detect pests and diseases are properly configured, optimized for speed, and thoroughly tested against known datasets to ensure their accuracy and reliability. - Functional Testing: Conducting unit tests, integration tests, and system tests to guarantee that every feature—like scanning, reminders, a mini-wiki, or expert consultation—functions as intended.
- Non-functional Testing: Assessing the system's performance, usability, and security to ensure it meets quality standards.
- Review of Documentation: Ensuring that all project documents (SRS, SPMP, design specs) are complete, consistent, and traceable back to the requirements.

Validation Activities

- Usability Testing: Involving real users, like small-scale palm owners, to see if the app is user-friendly, accessible, and easy to navigate, particularly for those who might not be tech-savvy. Accuracy of Results: Making sure that the AI-driven diagnoses for diseases and pests are clear, relevant, and actionable for users.
- Expert Feedback: Evaluating the quality and relevance of the expert advice offered through the app.
- System performance validation: Assessing how well the app performs in real-world scenarios, which includes:
- Response Time: Ensuring the app quickly processes image scans and provides results without delay.
- Resource Efficiency: Checking that the app operates smoothly on standard smartphones without draining the battery or hogging memory.
- Scalability: Testing how the system copes with multiple users or simultaneous consultations. Reliability: Making sure the app stays stable during long sessions and adapts to different network conditions.

6.3. Documentation

The Nekhlawi project follows the IEEE standards of software project management plans to guarantee that all the documentation is professional, consistent, and complete.

All the significant project-related paperwork and deliverables should be provided during the development process and at the year-end; here is the list of all these:

Problem Statement Report: The report that presents a clear articulation of the problem that the project is dealing with, the scope, and the high-level objectives of the solution that is being proposed.

Software Project Management Plan (SPMP): This is the general document of project management, schedule, budget, controls, and communication strategies.

Software Requirements Specification (SRS): This document is a formal document, and it contains all the functional and non-functional requirements that must be met by the finished software.

Software Design Specification (SDS): This document is used to outline the architecture, components, interfaces, and data design of the system, and this is a blueprint to be followed in implementation.

Final Report (Thesis): The entire and full process of the project's life cycle, starting with problem definition and literature review, but also including design, implementation, testing, and conclusion.

6.4. Quality Assurance

The team will form a quality assurance team specifically to make sure that the project is built upon a unified standards upon all members units and to make that it follows the tech industry quality standards like IEEE and ISO, in addition, the team will review the project upon the supervisors calls at any point and make corrections, additions accordingly.

6.5. Reviews and Audits

The team members will review their own progress and each other's work weekly both between members and supervisor. The supervisor feedback and notes is taken weekly to make modifications accordingly.

6.6. Problem Resolution

In each member task, if the member faces a problem, they must address it to the leader and the team members to help out with it. They will try to find a solution by analyzing or researching it or looking into similar problems and its solution, but if no solution is found they will discuss it to the supervisor and seek her guidance.

6.7. Process Improvement

This section outlines the strategies used to foster ongoing process improvement in the Nekhlawi project. The goal is to keep an eye on development efficiency, boost software quality, and ensure that everything aligns with international software engineering standards at every stage of the project.

6.7.1. Systems/Software Process Improvement Lead

The Systems/Software Process Improvement Lead plays a crucial role in consistently monitoring, evaluating, and enhancing the efficiency and quality of all software development efforts within the Nekhlawi project. This position involves analyzing performance metrics, pinpointing bottlenecks, and taking corrective measures to boost

productivity and minimize errors. The lead makes sure that lessons learned from each phase are documented and incorporated into future workflows. Key focus areas include refining coding standards, enhancing testing methods, and optimizing the accuracy of AI models. Regular feedback is gathered from developers, users, and supervisors to ensure a high level of system reliability, maintainability, and user satisfaction.

6.7.2. Systems Engineering Process Group

The Systems Engineering Process Group (SEPG) for the Nekhlawi project consists of the Project Manager, AI/ML Specialists, and members of the Quality Assurance team. This group is tasked with evaluating and improving engineering and management processes to ensure they meet IEEE and international standards. The SEPG holds review sessions after each major milestone to assess the project's technical practices, documentation quality, and process efficiency. Any identified weaknesses or inefficiencies are documented, and improvement actions are planned for the next development iterations. By maintaining thorough documentation, conducting risk assessments, and adhering to best practices, the SEPG fosters a structured and sustainable approach to continuous improvement and software process excellence.

REFERENCES

[1] Date palm, knowledge, skills, traditions and practices - UNESCO intangible cultural heritage," *Intangible Cultural Heritage*, https://ich.unesco.org/en/RL/date-palm-knowledge-skills-traditions-and-practices-01902 (accessed Sep. 29, 2025).

[2] KSA's World Heritage List - date palm, knowledge, skills, traditions and practices," *KSA for UNESCO*, https://ksaforunesco.org/heritages/date-palm-knowledge-skills-traditions-and-practices/ (accessed Sep. 29, 2025).

[3] A. I. Aldakhil et al., "Evaluating the economic impact of Integrated Pest Management (IPM) on Smallholder Date Palms Farmers - Journal of the Saudi Society of Agricultural Sciences," SpringerLink, https://link.springer.com/article/10.1007/s44447-025-00035-6 (accessed Sep. 29, 2025).

[4] FAO and AOAD, Value chain study – Date palm in the Arab region, Cairo: FAO, 2023. [Online]. Available: https://openknowledge.fao.org/server/api/core/bitstreams/ec3c70f9-b773-44de-9fc3-a5ae6a922cb8/content.

[5] "المركز الوطني للنخيل والتمور", Available: https://ncpd.gov.sa/en/news/32

[6] FAO and AOAD, Value chain study – Date palm in the Arab region, Cairo: FAO, 2023. [Online]. Available: https://openknowledge.fao.org/server/api/core/bitstreams/ec3c70f9-b773-44de-9fc3-a5ae6a922cb8/content .

[7] "FAO Knowledge Repository", Available: https://openknowledge.fao.org/items/056f87cf-29f6-4493-aa48-0312dbcbb127?.

[8] M. Alshehri and O. Alharbi, "Understanding the Landscape of Leveraging IoT for Sustainable Growth in Saudi Arabia," Jul. 2024, doi: 10.48550/arxiv.2407.04273. Available: https://arxiv.org/pdf/2407.04273.

[9] M. Abu-Zanona, S. Elaiwat, S. Younis, N. Innab, and M. M. Kamruzzaman, "Classification of Palm Trees Diseases using Convolution Neural Network," *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 6, Jan. 2022, doi: 10.14569/ijacsa.2022.01306111.

[10] M. Ahmed and A. Ahmed, "Palm tree disease detection and classification using residual network and transfer learning of inception ResNet," *PLoS ONE*, vol. 18, no. 3, p. e0282250, Mar. 2023, doi: 10.1371/journal.pone.0282250.

[11] M. Faisal, M. Alsulaiman, M. Arafah, and M. A. Mekhtiche, "IHDS: Intelligent Harvesting Decision System for Date Fruit Based on Maturity Stage Using Deep Learning and Computer Vision," IEEE Access, vol. 8, pp. 167985-167997, 2020, doi: 10.1109/ACCESS.2020.3023894.

[12] Faisal, M. *et al.* (2020) 'Deep learning and computer vision for estimating date fruits type, maturity level, and weight', *IEEE Access*, 8, pp. 206770–206782. doi:10.1109/access.2020.3037948.

[13] S. N. Nobel et al., "Palm Leaf Health Management: A Hybrid Approach for Automated Disease Detection and Therapy Enhancement.," IEEE Access, doi: 10.1109/access.2024.3351912.

[14] A. Magsi *et al.*, "A new hybrid algorithm for intelligent detection of sudden decline syndrome of date palm disease," *Scientific Reports*, vol. 13, no. 1, Sep. 2023, doi: 10.1038/s41598-023-41727-9.

[15] H. Alaa, K. Waleed, M. Samir, M. Tarek, H. Sobeah, and M. Abdul, "An Intelligent Approach for Detecting Palm Trees Diseases using Image Processing and Machine Learning," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 7, Jan. 2020, doi: 10.14569/ijacsa.2020.0110757.

[16] K. H. Arafat, "Advancements in Detecting Spot Fungal Diseases on Date Palm Leaves: Development and Validation of Chlorophyll-Based Detection Models," New Valley Journal of Agricultural Science, vol. 4, no. 2, pp. 16-36, 2024, doi: 10.21608/NVJAS.2024.287037.1286.

[17] K. H. Arafat, M. H. A.-R. Hassan, W. M. AlMetwaly, and W. H. Mohammed, "Early detection of date palm diseases in the New Valley Governorate using NDVI-Based Digital Cartographic modeling," *Al-Mağallah Al-ʿilmiyyaẗ Li ʿulūm Al-Rīyāḍaẗ Bi Ğamiʿāẗ Al-Minūfiyyaẗ*, vol. 5, no. 1, pp. 31–49, Mar. 2025, doi: 10.21608/nvjas.2025.365780.1314.

[18] M. Safran, W. Alrajhi, and S. Alfarhood, "DPXception: a lightweight CNN for image-based date palm species classification," *Frontiers in Plant Science*, vol. 14, Jan. 2024, Art. no. 1281724, doi:10.3389/fpls.2023.1281724.

[19] A. Hessane, A. E. Youssefi, Y. Farhaoui, B. Aghoutane, and F. Amounas, "A machine learning based framework for a Stage-Wise classification of date Palm White Scale disease," *Big Data Mining and Analytics*, vol. 6, no. 3, pp. 263–272, Apr. 2023, doi: 10.26599/bdma.2022.9020022.

[20] Y. Zarouit, H. Zekkouri, M. Ouhda, and B. Aksasse, "Date fruit detection dataset for automatic harvesting," *Data in Brief*, vol. 52, p. 109876, Nov. 2023, doi: 10.1016/j.dib.2023.109876.

[21] W. K. ElHelew, D. Abo Bakr, S. Zayan, and M. A. Mayhoub, "Classification of Dates Quality Using Deep Learning Technology Based on Captured Images," *Misr Journal of Agricultural Engineering*, vol. 41, no. 3, pp. 225-242, Jul. 2024, doi: 10.21608/mjae.2024.286079.1137.

[22] A. Almutairi et al., "Date Fruit Detection and Classification Based on Its Variety Using Deep Learning Technology," *IEEE Access*, vol. 12, pp. 190666-190677, 2024.

[23] "External Features-Based Approach to Date Grading and Analysis with Image Processing," vol. 6, no. 4, pp. 694–704, May 2022, doi: 10.28991/esj-2022-06-04-03. Available: https://doi.org/10.28991/esj-2022-06-04-03

[24] Esraa Hassan, Sarah Abu Ghazalah, Nora El-Rashidy, Tarek Abd El-Hafeez, and Mahmoud Y. Shams, "Sustainable deep vision systems for date fruit quality assessment using attention-enhanced deep learning models," *Frontiers in Plant Science*, vol. 16, Article 1521508, 2025, doi:10.3389/fpls.2025.1521508.

[25] A. Namoun, A. B. Alkhodre, A. A. Abi Sen, Y. Alsaawy, and H. Almoamari, "Dataset of Infected Date Palm Leaves for Palm Tree Disease Detection and Classification," *Data in Brief*, vol. 57, no. (2024), Art. no. 110933, Sep. 2024, doi:10.1016/j.dib.2024.110933.

[26] S. S. Harakannanavar, J. M. Rudagi, V. I. Puranikmath, A. Siddiqua, and R. Pramodhini, "Plant leaf disease detection using computer vision and machine learning algorithms," *Global Transitions Proceedings*, vol. 3, no. 1, pp. 305–310, Apr. 2022, doi: 10.1016/j.gltp.2022.03.016.

[27] S. A. A. Qadri, N.-F. Huang, T. M. Wani, and S. A. Bhat, "Advances and Challenges in Computer Vision for Image-Based Plant Disease Detection: A Comprehensive survey of machine and deep learning approaches," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 2639–2670, Apr. 2024, doi: 10.1109/tase.2024.3382731.

[28] Y. Tong, T.-H. Lee, and K.-S. Yen, "Deep Learning for Image-Based Plant Growth Monitoring: A Review," *International Journal of Engineering and Technology Innovation*, vol. 12, no. 3, pp. 225-246, 2022, doi:10.46604/ijeti.2022.8865.

[29] S. S. Harakannanavar, J. M. Rudagi, V. I. Puranikmath, A. Siddiqua, and R. Pramodhini, "Plant leaf disease detection using computer vision and machine learning algorithms," *Global Transitions Proceedings*, vol. 3, pp. 305-310, 2022, doi: 10.1016/j.gltp.2022.03.016.

[30] Sagheer, S. V. M., Orwel, P. V., Ameer, P. M., BaQais, A., & Kalathil, S. (2025). A Deep Learning Approach to Classification of Diseases in Date Palm Leaves. *Computers, Materials & Continua*.

[31] M. Culman, S. Delalieux, and K. Van Tricht, "Individual Palm Tree Detection Using Deep Learning on RGB Imagery to Support Tree Inventory," *Remote Sens.*, vol. 12, no. 21, p. 3476, Oct. 2020, doi: 10.3390/rs12213476.

[32] Cho, S., Kim, T., Jung, D. H., Park, S. H., Na, Y., Ihn, Y. S., & Kim, K. (2023). Plant growth information measurement based on object detection and image fusion using a smart farm robot. *Computers and Electronics in Agriculture, 207*, 107703.

[33] R. Nettle and J. Ingram, "Smart Farming Technologies and Changes to Farm Work: New Insights Into On-Farm Experiences," *Technological Forecasting & Social Change*, vol. 218, Art. no. 124227, 2025, doi:10.1016/j.techfore.2025.124227.

[34] G. Alkhaldi, A. Emam, and E. Mosbah, "An Economic Study: Date palm farmers in Al Ahsa Oasis-Kingdom of Saudi Arabia," *Universal Journal of Agricultural Research*, vol. 10, no. 6, pp. 794–803, Dec. 2022, doi: 10.13189/ujar.2022.100617.

[35] D. Homan and J. A. Du Preez, "Automated feature-specific tree species identification from natural images using deep semi-supervised learning," *Ecological Informatics*, vol. 66, p. 101475, Nov. 2021, doi: 10.1016/j.ecoinf.2021.101475.

[36] K. M. O. Nahar, N. Atitallah, M. M. A. Shquier, D. Z. Zreiqat, and K. M. Alhawiti, "Medicinal plant recognition based on the seedling image and deep learning," *Journal of Advanced Research in Applied Sciences and Engineering Technology*, pp. 209–219, Oct. 2024, doi: 10.37934/araset.58.1.209219.

[37] A. Ammar, A. Koubaa, and B. Benjdira, "Deep-Learning-Based Automated Palm Tree Counting and Geolocation in Large Farms from Aerial Geotagged Images," *Agronomy*, vol. 11, no. 8, p. 1458, Jul. 2021, doi: 10.3390/agronomy11081458.

[38] J. Xiong, D. Yu, S. Liu, L. Shu, X. Wang, and Z. Liu, "A Review of Plant Phenotypic Image Recognition Technology Based on Deep Learning," *Electronics*, vol. 10, no. 1, p. 81, Jan. 2021, doi: 10.3390/electronics10010081.

[39] A. Al-Abbad, M. Al-Jamal, Z. Al-Elaiw, F. Al-Shreed, and H. Belaifa, "A study on the economic feasibility of date palm cultivation in the Al-Hassa Oasis of Saudi Arabia," *Journal of Development and Agricultural Economics*, vol. 3, no. 9, pp. 463–468, Sep. 2018, doi: 10.5897/jdae.9000002.

[40] Saudi Vision 2030, "Overview of Vision Realization Programs," Vision 2030 Official Website. Available: https://www.vision2030.gov.sa/v2030/overview/.

[41] Saudi Vision 2030, "National Heritage Preservation Initiatives," Vision 2030 Official Website. Available: https://www.vision2030.gov.sa.